

Predicting porcini: a decade of sporocarp monitoring reveals the meteorological triggers of *Boletus edulis* fruiting in central European beech forests

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ABSTRACT

14 Despite recent molecular and genomic advances, the reproductive biology of ectomycorrhizal fungi
(EMF) remains poorly understood. In particular, the meteorological cues triggering sporocarp formation
16 remain largely unknown, limiting our ability to induce fruiting in the laboratory. To address this
knowledge gap, we analysed a decade-long (2015–2024) dataset of daily, near-exhaustive sporocarp
18 observations from an intensively monitored population of *Boletus edulis* in a central European beech
forest near Bielefeld, Germany. Using generalised linear mixed-effect models, we estimated the lagged
20 effects of temperature and precipitation on sporocarp formation and identified the conditions associated
with peak fruiting. Although sporocarps formed across a broad range of conditions, peak fruiting was
22 associated with a mean temperature of approximately 13°C averaged over the preceding 20 days and
increased linearly with precipitation accumulated over a 26-day window. Under projected warmer and
24 drier autumn conditions, *B. edulis* sporocarp formation in European beech forests is likely to decline.

26 Keywords: *Boletus edulis*; sporocarp formation; fruiting; climate

28 **Introduction**

30 *Boletus edulis* Bull., commonly known as the porcini, cèpe de Bordeaux, penny bun or Steinpilz, is a
 32 Basidiomycete that produces one of the world's most highly prized edible mushrooms. In parts of Europe,
 34 its economic value can even exceed that of timber, contributing substantially to local economic growth
 (Oria-de-Rueda et al., 2008). Ecologically, *B. edulis* forms ectomycorrhizal associations with a wide
 36 variety of host plants, from dominant European coniferous and deciduous tree species to Mediterranean
 and alpine shrubs (Alonso Ponce et al., 2011; Treindl and Leuchtman, 2019). In central Europe, it is
 38 commonly associated with European beech (*Fagus sylvatica*), the most widespread deciduous tree on the
 continent, which is increasingly threatened by drought, heat stress and other consequences of climate
 40 change (Diers et al., 2023; Leuschner et al., 2023; Martinez del Castillo et al., 2022; Obladen et al.,
 2021). Ectomycorrhizal fungi (EMF) such as *B. edulis* are thought to bolster tree resilience under such
 42 conditions by enhancing water and nutrient uptake, making a better understanding of the biology of EMF
 an urgent priority in the context of ongoing climate change. Owing to its economic and ecological
 44 significance, together with the relative ease of locating its fruiting bodies, *B. edulis* is emerging as a
 model system for studying EMF (Brejon Lamartinière et al., 2024; Hoffman et al., 2020; Santolamazza-
 Carbone et al., 2023; Tremble et al., 2023b, 2023a, 2020).

46 Despite advances in molecular genomics that have illuminated the chromosomal architecture, population
 structure and evolutionary history of *B. edulis* (Brejon Lamartinière et al., 2025, 2024; Tremble et al.,
 48 2023b, 2023a), fundamental aspects of its biology remain poorly understood. This gap partly arises due
 to our inability to cultivate this species through its complete life cycle under laboratory conditions, which
 50 precludes experimental manipulation. In particular, the environmental conditions that trigger sporocarp
 formation remain elusive, in contrast to many saprotrophic and cultivated fungi, where fruiting conditions
 52 have been characterised with remarkable precision under laboratory conditions (Guo and Xu, 2025). For

example, the optimal fruiting temperatures for producing multiple *Pleurotus* (oyster mushroom) species
54 have been determined to within a single degree Celsius (Guo and Xu, 2025).

56 In EMF, where fruiting has not yet been successfully induced in the laboratory, long-term field
monitoring of sporocarps has identified temperature and precipitation as the primary drivers of fruiting
58 (Andrew, 2025; Karavani et al., 2018; Martínez-Peña et al., 2012b, 2012a; Tsunoda et al., 2025).
However, these variables are not independent and may act jointly (Karavani et al., 2018), requiring their
60 joint inclusion in statistical models to disentangle their individual effects. Moreover, because sporocarp
development can take several days (Yang et al., 2025), the environmental conditions that initiate fruiting
62 are likely to act during the period preceding sporocarp emergence and observation. Consequently,
modelling sporocarp production requires accounting for the lagged effects of both temperature and
64 precipitation, while determining the length of these potentially distinct lag periods from daily-resolution
environmental data and sporocarp observations. To our knowledge, however, no study has modelled EMF
66 sporocarp production at daily resolution while jointly estimating distinct, separately optimised lag
periods for the effects of both temperature and precipitation.

68
In Bielefeld, north-west Germany, a population of *B. edulis* has been continuously monitored since 2015
70 (Hoffman et al., 2020). The study area, located west of the city, comprises multiple close woodland
patches dominated by *F. sylvatica*. Within these patches, sporocarp production has been continuously
72 tracked across 15–29 study sites (see methods for details), with every sporocarp sighting systematically
recorded. Using this long-term dataset, we investigated the meteorological conditions associated with
74 sporocarp formation in *B. edulis*. Specifically, we first jointly estimated the optimal lag periods for
temperature and precipitation that best predict sporocarp formation using generalised mixed effect
76 models (GLMMs). We then used the model with the best fitting lag periods to estimate the optimal
meteorological conditions leading to *B. edulis* sporocarp formation.

78 **Materials and Methods**

Study population and fieldwork

80 From 2015 to 2024 inclusive, we monitored *B. edulis* sporocarp production across our study area near
Bielefeld (52°03'42.2"N 8°27'03.3"E). The area comprises multiple woodland patches dominated by *F.*
82 *sylvatica* followed by common oak (*Quercus robur*) and birch (*Betula pendula*). Monitoring initially
focused on 15 study sites, with this number gradually increasing to 29 as additional fruiting locations
84 were discovered, although some earlier sites have since ceased fruiting. Site area ranged from 30 m² to
approximately 1.2 ha (mean = 2,340 m², s.d. = 4,295 m²). Sporocarp production was monitored at every
86 active site, with visits conducted 2–4 times a week from early August until the first heavy frost or,
alternatively, until approximately two weeks after the last sporocarp was sighted, whichever occurred
88 first (typically around mid-October to mid-November). During each visit, the date and location of every
sporocarp was recorded using a handheld GPS device.

90

Environmental data

92 To investigate the meteorological conditions that trigger sporocarp formation, we retrieved the 24h mean
temperature (C) and precipitation (mm) for our study area using Copernicus E-OBS *in situ*
94 meteorological data observations at the maximum resolution of 0.1 degrees (Cornes et al., 2018). As all
of our study sites fitted within a single grid cell, we extracted the environmental data corresponding to
96 the centroid of the study area using bilinear interpolation, which computes a distance-weighted average
from the four nearest cell centres to better approximate local conditions. For subsequent analyses, we
98 considered the period between August 5th—the date of our earliest sporocarp sighting—and November
21st, the date of our latest sporocarp sighting. All downstream analyses were conducted using sporocarp
100 data pooled from all of the sites to maximise statistical power. We additionally repeated the analyses
separately for the two most productive sites to explore inter-site variation while retaining adequate
102 statistical power.

Data analyses

1. Lag periods of temperature and precipitation effects

We first conducted an exploratory analysis to identify the appropriate lag periods for the effects of temperature and precipitation on sporocarp formation across the study period. Daily mean temperature and precipitation values were averaged over right-aligned sliding windows of lengths 2, 5, 8, 11, 14, 17, 20, 23, 26, 29, 32 and 35 days. To determine the most appropriate lag periods for both variables, we constructed separate GLMMs predicting daily sporocarp counts as a function of all possible combinations of temperature and precipitation window lengths. These models were specified with a negative binomial error distribution to account for overdispersion and the large number of zero observations (sporocarps were not sighted on 85.8% of days), and included year as a random effect to account for inter-annual variation. Temperature was modelled as a quadratic term to allow for a non-linear relationship with sporocarp production as documented in saprotrophic fungi (Guo and Xu, 2025), while precipitation was modelled as a linear predictor because no upper threshold has yet been identified beyond which additional precipitation no longer promotes sporocarp formation (Tsunoda et al., 2025). The optimal lag periods for temperature and precipitation were then identified by selecting their corresponding window lengths in the best performing model based on Akaike's Information Criterion (AIC). These models were implemented using the GLMMTMB package (version 1.1.13, Brooks et al., 2017).

2. Optimal conditions triggering sporocarp formation

Using the mean temperature and precipitation values computed over the optimal window lengths identified above, we performed two-dimensional kernel density estimation to compare the meteorological conditions associated with sporocarp formation against the full range of meteorological conditions observed during throughout the study period. This comparison allowed us to investigate whether sporocarp occurrence reflects specific climatic preferences, or whether fruiting simply occurs in proportion to the frequency of the prevailing meteorological conditions, such that sporocarps are more

128 common under frequently occurring conditions rather than under biologically preferred ones. Kernel
density estimates were computed using the MASS package (version 7.3-65, Venables and Ripley, 2002)
130 and density surfaces were visualised with GGPLOT2 (version 3.5.2, Wickham 2016). Finally, we
formally derived the optimal meteorological conditions for sporocarp formation using the best-fitting
132 GLMM described above. All analyses and visualisations were implemented in R v4.5.1 (R core team
2025). To maximise transparency and reproducibility, the data and code used to analyze them have been
134 archived at zenodo (<https://zenodo.org/records/20591123>).

136 Results and discussion

Across the full dataset, the interval between the earliest sporocarp sighting recorded across all years and
 138 the latest sporocarp sighting across all years was 86 days. During this overall fruiting window, 24h mean
 temperatures ranged from -2.5 to 25.9 °C (FIG 1A), while daily mean precipitation varied between zero
 140 and 42.6 mm (FIG. 1B). The total number of sporocarps varied substantially among years, ranging from
 four in 2016 to 354 in 2020 (FIG 1). In several years, fruiting was not continuous but occurred in multiple
 142 distinct pulses within a season (e.g. 2020; Fig. 1).

144 Sites were visited frequently (2–4 times per week) and locations known from previous observations to
 produce sporocarps were consistently revisited, minimising the chance of missed fruiting events. This
 146 intensive, targeted monitoring approach made the dataset as exhaustive as practically achievable. The
 near absence of sporocarps in 2016 therefore appears to be a genuine biological signal rather than a
 148 sampling artefact, likely reflecting exceptionally dry conditions during the peak fruiting period (FIG 1B).
 This interpretation is further supported by the occurrence of moderate fruiting in the preceding and
 150 following years (FIG 1). Although total sporocarp counts were generally higher in the latter half of the
 study, partly because new fruiting sites were discovered over time, no consistent trends in temperature
 152 or precipitation were observed across the years. We therefore interpret variation in sporocarp numbers
 among years as primarily reflecting true biological variation rather than sampling artifacts.

154
 To determine the most appropriate lag periods for the effects of temperature and precipitation on
 156 sporocarp production, we constructed separate GLMMs predicting daily sporocarp counts as a function
 of all possible combinations of window length for both variables (see Methods for details). The best
 158 fitting model identified an optimal lag period of 20 days for temperature, both when all sites were
 analysed jointly, as well as when the two most productive sites were modelled separately (Table S1, Fig.
 160 2, Fig. 3). For precipitation, the best fitting model identified a lag period of 26 days when sites were

combined, and 26-32 days when the two largest sites were analysed separately (Table S1, Fig. 2, Fig. 3).

162 These findings indicate that sporocarp formation is triggered most strongly by approximately three weeks of favourable temperature conditions, while the effect of precipitation has a longer lag period, likely

164 reflecting the time needed to build and up and maintain elevated soil moisture levels (Karavani et al., 2018). In line with this, previous studies incorporating both variables similarly reported precipitation

166 lags of approximately one month across multiple EMF species (Karavani et al., 2018), whereas studies excluding temperature yielded shorter estimated lags of less than two weeks for the effects of

168 precipitation (Tsunoda et al., 2025).

170 To evaluate the meteorological conditions triggering sporocarp formation, we applied two-dimensional kernel density estimation to mean temperature and precipitation values computed over right-aligned

172 sliding windows corresponding to the previously identified optimal lag periods. This analysis was first conducted across all days of the study period to characterise the full range of environmental conditions

174 encountered during monitoring. It was then repeated only using days on which sporocarps were observed to distinguish whether the occurrence of sporocarps reflects specific climatic preferences or whether

176 fruiting simply occurs in proportion to the frequency of the prevailing meteorological conditions. Across the full study period, meteorological conditions spanned a broad range of temperatures (20 day means:

178 5–20 °C) while precipitation was more constrained, with mean daily values over 35-day windows rarely falling below 1mm or exceeding 4mm (Fig. 4A).

180

By contrast, fruiting events were concentrated within a narrower range of conditions, particularly

182 temperatures between 10 and 15 °C and precipitation levels between 2 and 4mm per day. Furthermore, fruiting was largely absent under the temperature conditions most frequently observed during the study

184 period (5–10°C). This pattern was consistent both when all sites were analysed together and when the two most productive sites were analysed separately. Notably, two apparent temperature optima were

associated with higher sporocarp production (Fig. 4B), a pattern that was also present when sites A and B were analysed separately (Fig. 4C, D). However, these apparent peaks were separated by intermediate temperature and precipitations that were rarely observed during the study period (Fig. 4A), potentially obscuring a continuous underlying response surface. Beyond these apparent peaks, there was no evidence that the effect of temperature depended on precipitation or *vice versa*, arguing against the inclusion of an interaction term in the subsequent models.

To formally quantify relationships between the meteorological variables and sporocarp formation while accounting for repeated observations across years and sites, we used the best fitting GLMMs: one including all sites and two additional models fitted respectively for the two most productive sites. Precipitation had a consistently positive effect on sporocarp formation across all models, in agreement with the results of the two-dimensional kernel density estimation and the general expectation that higher rainfall should promote sporocarp formation (Fig. 4B4E). This finding is also consistent with previous studies of other EMF, which likewise found no evidence for an upper precipitation threshold beyond which fruiting declines (Tsunoda et al., 2025).

By contrast, temperature exhibited a significant quadratic relationship with sporocarp production, revealing an optimum at approximately 13 °C (Fig. 4F) that differed by less than 0.6 °C among the three models (Table S1). This predicted optimum is similar to the temperature used to trigger sporocarp formation in multiple species of cultivated saprotrophic fungi including *Pleurotus*, *Morchella* and *Hypsizygus* spp. (Guo and Xu, 2025). Because these taxa span distantly related fungal families with distinct ecologies, this similarity suggests that the regulatory pathways governing temperature-dependent sporocarp formation may be broadly conserved across mushroom-forming fungi. This convergence is perhaps not unexpected given morphological similarities among the sporocarps of mycorrhizal and saprotrophic basidiomycetes.

212 Conclusion

214 In this study, we identified the meteorological conditions most strongly associated with *B. edulis* sporocarp formation in a beech woodland near Bielefeld, Germany. Temperature emerged as the main short-term predictor, with a 20-day mean temperature of approximately 13°C being associated with the highest probability of sporocarp formation. By contrast, the effect of precipitation was not immediate but accumulated over a longer period, with elevated rainfall over the preceding 26 days increasing the likelihood of fruiting. In the context of environmental change, meteorological conditions in many regions are expected to shift away from the temperature ranges currently associated with optimal sporocarp production in many saprotrophic fungi (Guo and Xu, 2025). Our results suggest that a similar change may also occur for *B. edulis* populations across much of central Europe. However, occasional sightings of sporocarps at higher temperatures in our study suggest that some capacity for persistence may exist, whether mediated by phenotypic plasticity or evolutionary adaptation. Long-term monitoring across broader climatic gradients will be required to determine the extent to which such responses can buffer future declines in fruiting.

226

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230 References

- Alonso Ponce, R., Águeda, B., Ágreda, T., Modrego, M.P., Aldea, J., Fernández-toirán, L.M., Martínez-peña, F., 2011. Rockroses and *Boletus edulis* ectomycorrhizal association: realized niche and climatic suitability in Spain. *Fungal Ecology* 4, 224–232. <https://doi.org/10.1016/j.funeco.2010.10.002>
- Andrew, C., 2025. Not always optimal: Fungal fruiting triggers indicate climate sensitivity in cooler regions. *Fungal Ecology* 75, 101416. <https://doi.org/10.1016/j.funeco.2025.101416>
- Brejon Lamartinière, E., Tremble, K., Dentinger, B.T.M., Dasmahapatra, K.K., Hoffman, J.I., 2025. A haplotype-resolved chromosomal reference genome for the porcini mushroom *Boletus edulis*. *G3 Genes|Genomes|Genetics* jkaf069. <https://doi.org/10.1093/g3journal/jkaf069>
- Brejon Lamartinière, E., Tremble, K., Dentinger, B.T.M., Dasmahapatra, K.K., Hoffman, J.I., 2024. Runs of homozygosity reveal contrasting histories of inbreeding across global lineages of the edible porcini mushroom, *Boletus edulis*. *Molecular Ecology* 33, e17470. <https://doi.org/10.1111/mec.17470>
- Brooks, M.E., Kristensen, K., van Benthem, K.J., Magnusson, A., Berg, C.W., Nielsen, A., Skaug, H.J., Machler, M., Bolker, B.M., 2017. glmmTMB balances speed and flexibility among packages for zero-inflated generalized linear mixed modeling. *The R journal* 9, 378–400. <https://doi.org/10.3929/ethz-b-000240890>
- Cornes, R.C., van der Schrier, G., van den Besselaar, E.J.M., Jones, P.D., 2018. An Ensemble Version of the E-OBS Temperature and Precipitation Data Sets. *Journal of Geophysical Research: Atmospheres* 123, 9391–9409. <https://doi.org/10.1029/2017JD028200>
- Diers, M., Weigel, R., Leuschner, C., 2023. Both climate sensitivity and growth trend of European beech decrease in the North German Lowlands, while Scots pine still thrives, despite growing sensitivity. *Trees* 37, 523–543. <https://doi.org/10.1007/s00468-022-02369-y>
- Guo, X., Xu, J., 2025. Impacts of global climate change on mushroom production: challenges and opportunities. *Agriculture Communications* 3, 100091. <https://doi.org/10.1016/j.agrcom.2025.100091>
- Hoffman, J.I., Nagel, R., Litzke, V., Wells, D.A., Amos, W., 2020. Genetic analysis of *Boletus edulis* suggests that intra-specific competition may reduce local genetic diversity as a woodland ages. *Royal Society Open Science* 7. <https://doi.org/10.1098/RSOS.200419>
- Karavani, A., De Cáceres, M., Martínez de Aragón, J., Bonet, J.A., de-Miguel, S., 2018. Effect of climatic and soil moisture conditions on mushroom productivity and related ecosystem services in Mediterranean pine stands facing climate change. *Agricultural and Forest Meteorology* 248, 432–440. <https://doi.org/10.1016/j.agrformet.2017.10.024>
- Leuschner, C., Weithmann, G., Bat-Enerel, B., Weigel, R., 2023. The Future of European Beech in Northern Germany—Climate Change Vulnerability and Adaptation Potential. *Forests* 14, 1448. <https://doi.org/10.3390/f14071448>
- Martinez del Castillo, E., Zang, C.S., Buras, A., Hacket-Pain, A., Esper, J., Serrano-Notivoli, R., Hartl, C., Weigel, R., Klesse, S., Resco de Dios, V., Scharnweber, T., Dorado-Liñán, I., van der Maaten-Theunissen, M., van der Maaten, E., Jump, A., Mikac, S., Banzragch, B.-E., Beck, W., Cavin, L., Claessens, H., Čada, V., Čufar, K., Dulamsuren, C., Gričar, J., Gil-Pelegrín, E., Janda, P., Kazimirovic, M., Kreyling, J., Latte, N., Leuschner, C., Longares, L.A., Menzel, A., Merela, M., Motta, R., Muffler, L., Nola, P., Petritan, A.M., Petritan, I.C., Prislan, P., Rubio-Cuadrado, Á., Rydval, M., Stajić, B., Svoboda, M., Toromani, E., Trotsiuk, V., Wilmking, M., Zlatanov, T., de

- Luis, M., 2022. Climate-change-driven growth decline of European beech forests. *Commun Biol* 5, 1–9. <https://doi.org/10.1038/s42003-022-03107-3>
- Martínez-Peña, F., Ágreda, T., Águeda, B., Ortega-Martínez, P., Fernández-Toirán, L.M., 2012a. Edible sporocarp production by age class in a Scots pine stand in Northern Spain. *Mycorrhiza* 22, 167–174. <https://doi.org/10.1007/s00572-011-0389-8>
- Martínez-Peña, F., de-Miguel, S., Pukkala, T., Bonet, J.A., Ortega-Martínez, P., Aldea, J., Martínez De Aragón, J., 2012b. Yield models for ectomycorrhizal mushrooms in *Pinus sylvestris* forests with special focus on *Boletus edulis* and *Lactarius group deliciosus*. *Forest Ecology and Management* 282, 63–69. <https://doi.org/10.1016/j.foreco.2012.06.034>
- Obladen, N., Dechering, P., Skiadaresis, G., Tegel, W., Keßler, J., Höllerl, S., Kaps, S., Hertel, M., Dulamsuren, C., Seifert, T., Hirsch, M., Seim, A., 2021. Tree mortality of European beech and Norway spruce induced by 2018-2019 hot droughts in central Germany. *Agricultural and Forest Meteorology* 307, 108482. <https://doi.org/10.1016/j.agrformet.2021.108482>
- Oria-de-Rueda, J.A., Martín-Pinto, P., Olaizola, J., 2008. Bolete Productivity of Cistaceous Scrublands in Northwestern Spain1. *Econ Bot* 62, 323–330. <https://doi.org/10.1007/s12231-008-9031-x>
- R Core Team, 2025. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>.
- Santolamazza-Carbone, S., Iglesias-Bernabé, L., Landin, M., Rueda, E.B., Barreal, M.E., Gallego, P.P., 2023. Artificial intelligence unveils key interactions between soil properties and climate factors on *Boletus edulis* and *B. reticulatus* mycelium in chestnut orchards of different ages. *Front. Soil Sci.* 3. <https://doi.org/10.3389/fsoil.2023.1159793>
- Treindl, A.D., Leuchtmann, A., 2019. A king amongst dwarfs: *Boletus edulis* forms ectomycorrhiza with dwarf willow in the Swiss Alps. *Alp Botany* 129, 185–189. <https://doi.org/10.1007/s00035-019-00218-2>
- Tremble, K., Brejon Lamartinière, E., Bradshaw, A.J., Arzú, R.F., Hoffman, J., Dentinger, B.T.M., 2023a. From populations to pan-genomes: investigating the role of ecology and genomic architecture in maintaining species boundaries in the porcini mushroom, *Boletus edulis*. <https://doi.org/10.1101/2023.09.05.556370>
- Tremble, K., Hoffman, J.I., Dentinger, B.T.M., 2023b. Contrasting continental patterns of adaptive population divergence in the holarctic ectomycorrhizal fungus *Boletus edulis*. *New Phytologist* 237, 295–309. <https://doi.org/10.1111/nph.18521>
- Tremble, K., Suz, L.M., Dentinger, B.T.M., 2020. Lost in translation: Population genomics and long-read sequencing reveals relaxation of concerted evolution of the ribosomal DNA cistron. *Molecular Phylogenetics and Evolution* 148, 106804. <https://doi.org/10.1016/j.ympev.2020.106804>
- Tsunoda, Y., Narita, A., Kitamura, K., Gisusi, S., 2025. The cumulative effects of precipitation determine the occurrence of *Tricholoma matsutake* fruiting bodies. *Fungal Ecology* 75, 101415. <https://doi.org/10.1016/j.funeco.2025.101415>
- Venables, B., Ripley, B., 2002. Modern Applied Statistics With S, in: Springer. <https://doi.org/10.1007/b97626>
- Wickham, H., 2016. ggplot2: Elegant Graphics for Data Analysis. Springer-Verlag, New York.
- Yang, T., Mu, H., Dai, L., Liu, J., Xu, X., Gao, F., Fang, Y., Jian, S., He, M., Zhang, C., 2025. First Report on the Artificial Cultivation Techniques of *Buchwaldoboletus xylophilus* (Boletales, Boletaceae, *Buchwaldoboletus*) in Southwest China. *Journal of Fungi* 11. <https://doi.org/10.3390/jof11030172>
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Data availability statement

236 All the data and code used to produce the analysis and the figures have been archived at
Zotero (<https://zenodo.org/records/20591123>) and Github
238 (https://github.com/ebregonl/Boletus_fruiting_triggers).

240 Figure. 1.

Overview of within- and among-year variation in meteorological conditions and sporocarp observations
 242 across the ten-year study period. Panel A shows 24h mean temperature (°C) values above the x axis (in
 red), while panel (B) shows 24h mean precipitation (mm) values above the x axis (in blue). Daily
 244 sporocarp sightings are plotted below the x-axes of both panels (in black). On the right side of each panel,
 the annual distribution of 24h mean temperature (in red) and precipitation (in blue) values are represented
 246 with using simplified boxplots (top), summarizing the first three quartiles interquartile range (box) and
 the most extreme values within 1.5 times the interquartile range (whiskers), and density plots (bottom),
 248 summarising the frequency of observations through a based on single-dimension kernel density
 estimation. Grey shading areas highlights indicate the interval between the first and last sporocarp
 250 sighting of each season. In 2016, sporocarps were only sighted on one day.

Figure 2.

252 Heatmap of the model fit sensitivity to the sliding window lengths (lag time) used to summarise
temperature (y axis) and precipitation (x axis) preceding sporocarp observations. The three panels
254 represent, left to right, the models fitted with the data from all sites combined, the largest site (site A),
and the second largest site (site B). The fill represents the Akaike Information Criterion (AIC), scaled to
256 be comparable among the three panels despite sample sizes differences and reversed such that darker red
indicates better relative fit.

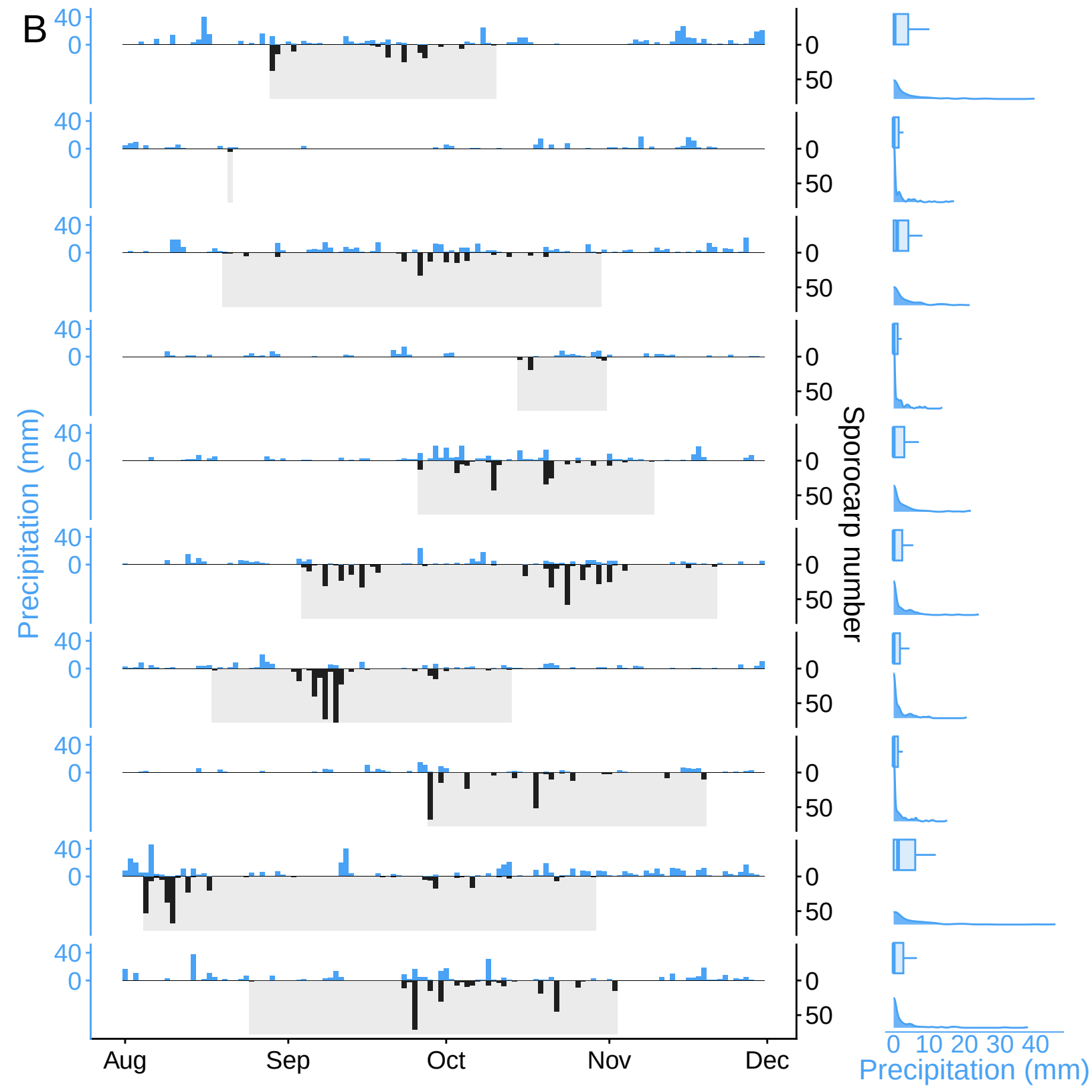
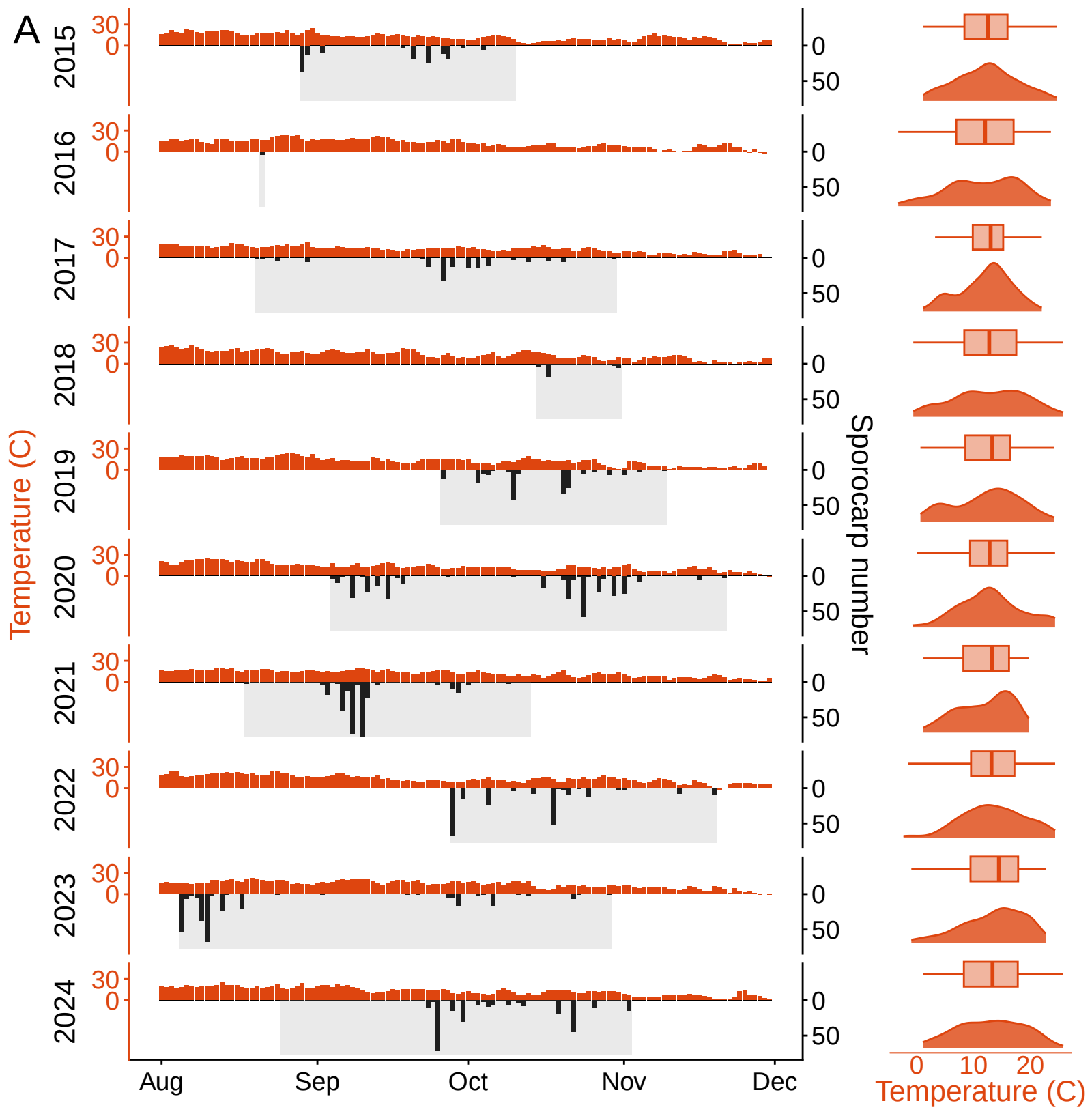
258 Figure. 3

260 Δ AIC values across sliding-window lengths when the complementary window is held constant at optimum. A) Precipitation window fixed at 26 days, with the temperature window length varying from 2 to 35 days (x axis). B) Temperature window length fixed at 20 days, with the precipitation window length varying from 2 to 35 days (x axis). Δ AIC represents the difference in AIC between each model and the best fitting model for that grouping, such that lower Δ AIC values indicate better model fit. The lines 264 represent the models fitted with the data from all sites combined in red, the most productive site in green and the second most productive site in blue.

266 Figure. 4

268 First row: two-dimensional kernel density estimates of the 20-day mean temperature and 26-day mean precipitation of all days of the monitored period A), fruiting days only across all sites combined B), fruiting days only at the most productive site C), and fruiting days only at the second most productive site D). Densities were rescaled to between 0 and 1 within each panel to simplify cross-panel comparison.

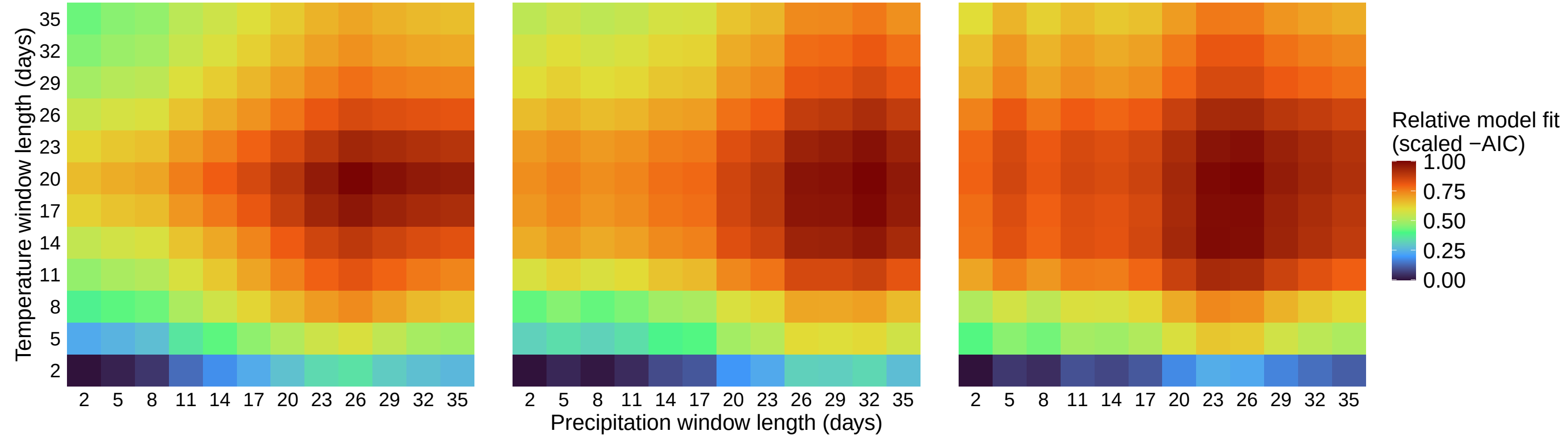
272 Second row: Predicted relative sporocarp production as a function of E) the 26-day mean precipitation with temperature held constant at the study mean, and F) the 20-day mean temperature with precipitation held constant at the study mean from the negative binomial generalised linear mixed effect model. The lines represent the models fitted with the data from all sites combined in red, the most productive site in green and the second most productive site in blue. The dashed vertical line represents the optimal temperature across all sites, 13.7 C.



All sites

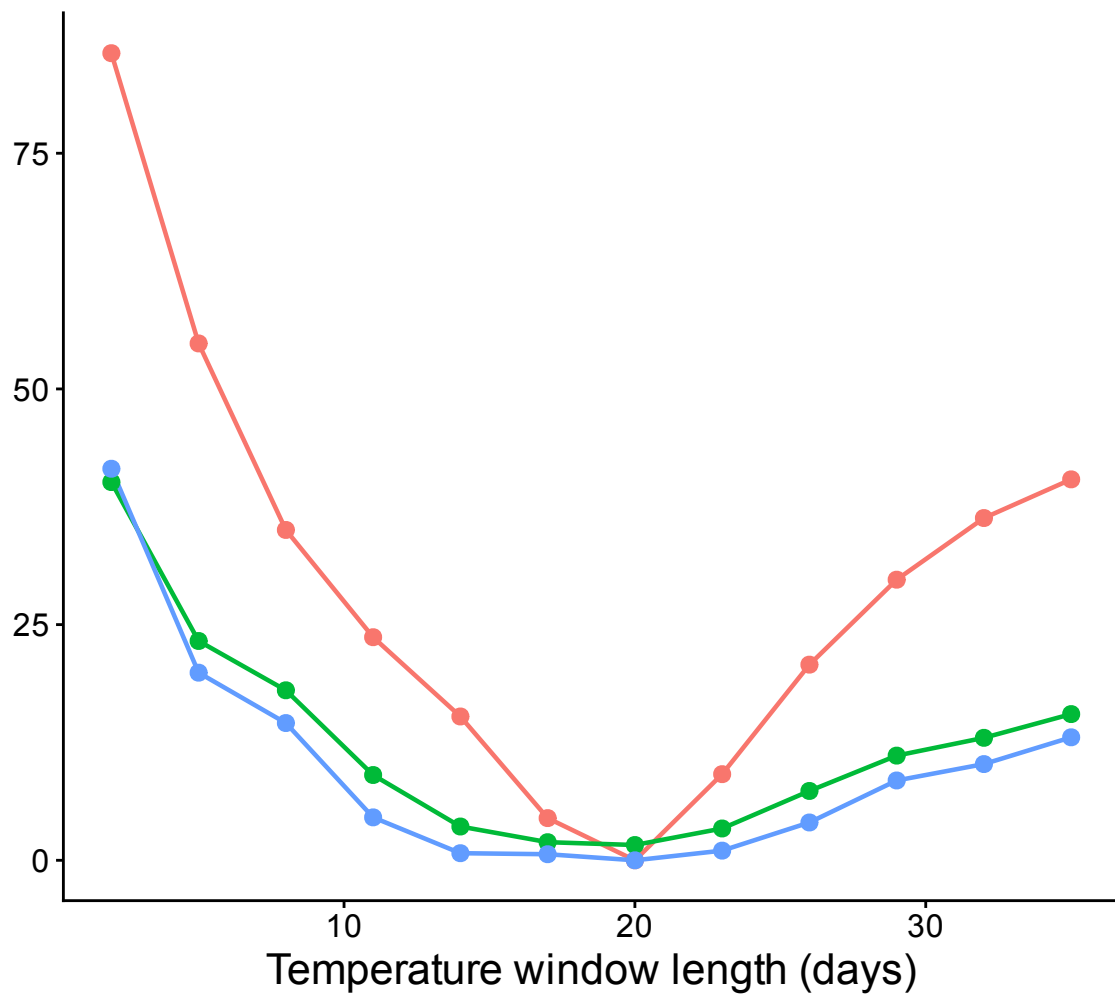
Site A

Site B



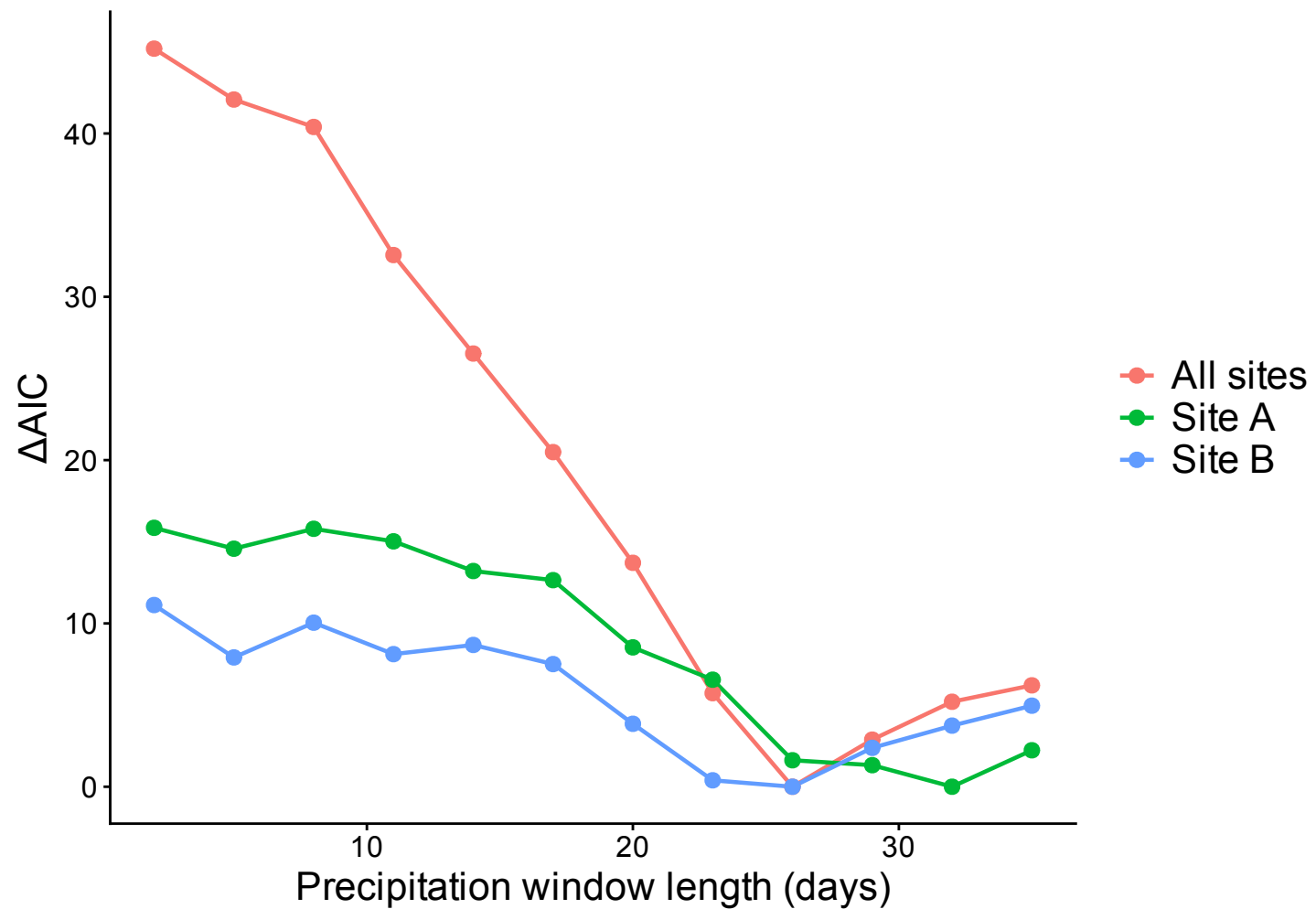
A

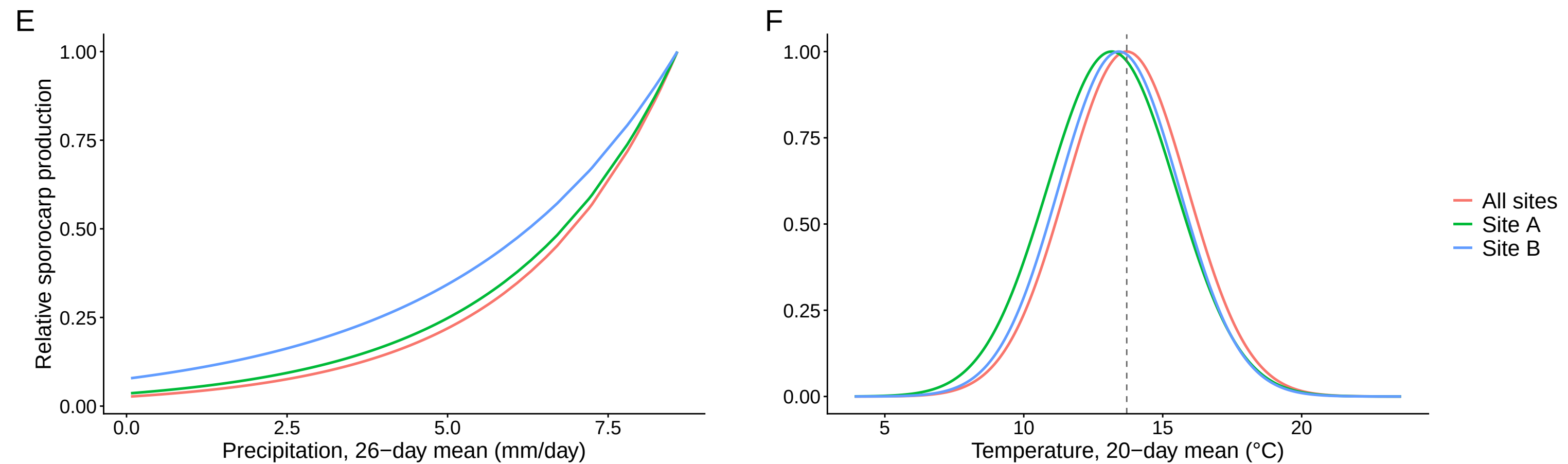
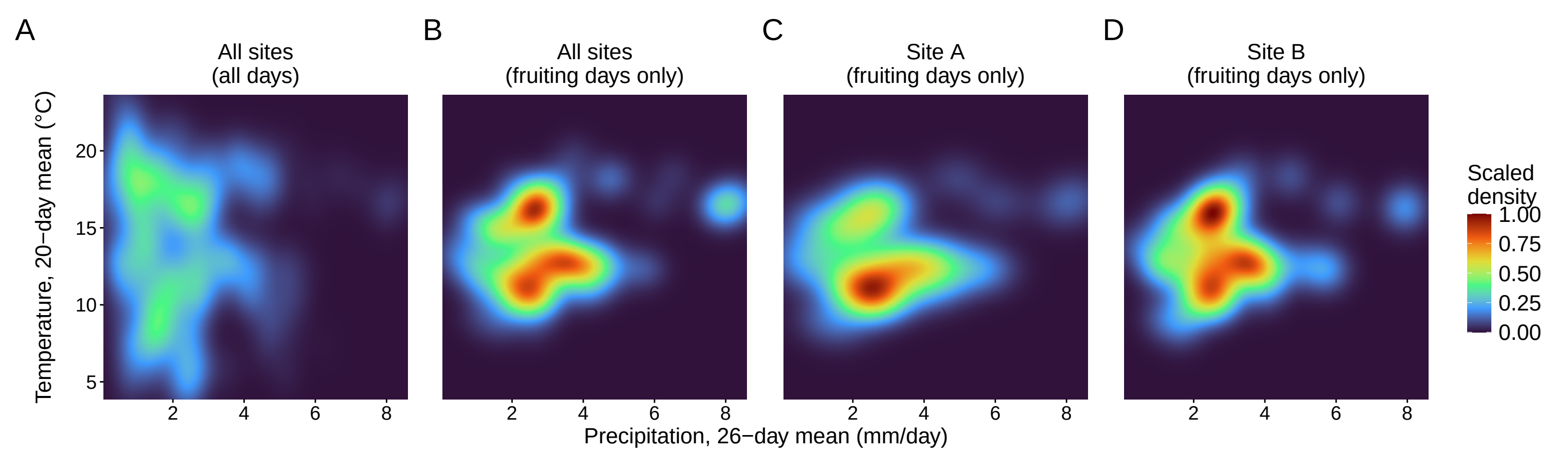
Precipitation window fixed at 26 days



B

Temperature window fixed at 20 days





Supplementary for the manuscript:

Predicting porcini: a decade of sporocarp monitoring reveals the meteorological triggers of *Boletus edulis* fruiting in central European beech forests

Etienne Brejon Lamartiniere and Joseph Ivan Hoffman

Table S1: Sensitivity of model fit and parameter estimates to the temperature and precipitation window lengths.

Each row is one combination of temperature (win_T) and precipitation (win_P) right-aligned sliding-window lengths (days), used to summarise the meteorological conditions preceding daily sporocarp counts. For each combination, a separate model was fitted with the data from all sites and from the two most productive sites (site A, site B) separately. The columns AIC and dAIC represent the Akaike's Information Criterion and the difference from the best-fitting model within each site (best model = 0). The columns coef_P, se_P and pval_P give the estimate, standard error and p-value for the effect of precipitation. Since Temperature was fitted as a quadratic term, it comprises a linear term (T1) and a quadratic term (T2). Therefore, columns coef_T1, se_T1, pval_T1 and coef_T2, se_T2, pval_T2 give the estimate, standard error and p-value of the linear term and the quadratic term, respectively. Finally, peak_T represents the model-predicted optimal temperature in Celsius. All values are rounded to five decimals.

win T	win P	AIC	coef P	se P	pval P	coef T1	se T1	pval T1	coef T2	se T2	pval T2	peak T	site	dAIC
2	2	1904.82865	0.02179	0.01893	0.2497	0.68017	0.11816	0	-0.02649	0.00452	0	12.84049	All	131.99089
5	2	1874.23154	0.03061	0.0187	0.1016	1.13217	0.16639	0	-0.0431	0.00623	0	13.13329	All	101.39378
8	2	1854.06773	0.03255	0.01873	0.0823	1.47431	0.20388	0	-0.05616	0.00759	0	13.12626	All	81.22997
11	2	1842.71958	0.02944	0.01897	0.12068	1.71976	0.23045	0	-0.06531	0.00852	0	13.16655	All	69.88182
14	2	1833.41199	0.02647	0.01895	0.16249	1.97131	0.25553	0	-0.07431	0.00937	0	13.26494	All	60.57423
17	2	1822.97558	0.02341	0.0189	0.21549	2.29856	0.29057	0	-0.08539	0.0105	0	13.4593	All	50.13782
20	2	1818.03329	0.02057	0.01901	0.27913	2.58993	0.32221	0	-0.0947	0.01147	0	13.67377	All	45.19552
23	2	1823.81716	0.02066	0.01904	0.27803	2.65613	0.32934	0	-0.09618	0.01162	0	13.80845	All	50.97939
26	2	1832.18741	0.02298	0.01914	0.22981	2.65942	0.32983	0	-0.09536	0.01154	0	13.94447	All	59.34965
29	2	1839.81994	0.02504	0.01921	0.19239	2.6882	0.33367	0	-0.09536	0.01157	0	14.09444	All	66.98218
32	2	1845.36726	0.02578	0.01925	0.18047	2.73618	0.33922	0	-0.09637	0.01168	0	14.19686	All	72.5295
35	2	1848.60829	0.02595	0.0192	0.17639	2.82717	0.34952	0	-0.09891	0.01196	0	14.2915	All	75.77053
2	5	1901.58856	0.05918	0.02677	0.02704	0.66572	0.11812	0	-0.02584	0.00452	0	12.88202	All	128.75079
5	5	1871.54787	0.06331	0.02667	0.01759	1.11634	0.16727	0	-0.04244	0.00626	0	13.1529	All	98.7101
8	5	1850.24842	0.07196	0.02661	0.00684	1.45801	0.20411	0	-0.05554	0.00759	0	13.12685	All	77.41066
11	5	1838.61729	0.07113	0.02693	0.00826	1.69971	0.23007	0	-0.06458	0.0085	0	13.16036	All	65.77952
14	5	1829.94403	0.06505	0.02707	0.01627	1.9408	0.25435	0	-0.07318	0.00932	0	13.25971	All	57.10626
17	5	1819.7114	0.06173	0.02731	0.02377	2.26624	0.28984	0	-0.0842	0.01047	0	13.45761	All	46.87363
20	5	1814.91731	0.05883	0.02754	0.03268	2.55719	0.32193	0	-0.09351	0.01146	0	13.67384	All	42.07955
23	5	1820.79421	0.05895	0.02794	0.03485	2.61929	0.32871	0	-0.09485	0.0116	0	13.80792	All	47.95644
26	5	1828.91128	0.06303	0.02816	0.0252	2.61776	0.32837	0	-0.09388	0.01149	0	13.9414	All	56.07352
29	5	1836.1592	0.06783	0.02842	0.017	2.64116	0.33121	0	-0.09371	0.01148	0	14.09148	All	63.32144
32	5	1841.44367	0.07021	0.02847	0.01365	2.68774	0.33619	0	-0.09468	0.01158	0	14.19363	All	68.6059
35	5	1844.56477	0.07117	0.02847	0.01242	2.77587	0.34598	0	-0.09716	0.01184	0	14.28483	All	71.727
2	8	1897.47956	0.1016	0.03259	0.00182	0.66924	0.11849	0	-0.0259	0.00453	0	12.92146	All	124.6418
5	8	1868.42979	0.09892	0.03236	0.00224	1.12572	0.16912	0	-0.04267	0.00633	0	13.19118	All	95.59203
8	8	1848.52399	0.09936	0.03242	0.00218	1.46382	0.20643	0	-0.05559	0.00767	0	13.16677	All	75.68623
11	8	1836.60776	0.10066	0.03294	0.00224	1.69911	0.23136	0	-0.06442	0.00854	0	13.18856	All	63.77
14	8	1828.12346	0.09273	0.03306	0.00503	1.93202	0.25475	0	-0.07275	0.00933	0	13.27831	All	55.28569
17	8	1818.14999	0.08792	0.03348	0.00863	2.25151	0.2897	0	-0.08356	0.01046	0	13.47176	All	45.31223
20	8	1813.2374	0.08633	0.03389	0.01086	2.54281	0.3219	0	-0.09289	0.01146	0	13.68657	All	40.39964
23	8	1819.2371	0.08602	0.03447	0.01258	2.60266	0.32885	0	-0.09416	0.0116	0	13.81992	All	46.39933
26	8	1827.56716	0.08909	0.03485	0.01057	2.5973	0.32857	0	-0.09307	0.01149	0	13.95333	All	54.72939
29	8	1834.82308	0.09459	0.03522	0.00723	2.61481	0.33106	0	-0.09272	0.01147	0	14.1012	All	61.98531
32	8	1839.92259	0.09932	0.03556	0.00523	2.65477	0.33538	0	-0.09346	0.01155	0	14.20318	All	67.08483
35	8	1842.92732	0.10081	0.03538	0.00438	2.73926	0.34474	0	-0.09582	0.0118	0	14.29412	All	70.08955
2	11	1886.93607	0.17401	0.03705	0	0.67321	0.11914	0	-0.02605	0.00455	0	12.91946	All	114.09831

5	11	1857.22966	0.1726	0.03621	0	1.14454	0.17166	0	-0.04333	0.00642	0	13.20814	All	84.3919
8	11	1838.30991	0.1686	0.03632	0	1.4903	0.21072	0	-0.05651	0.00782	0	13.18619	All	65.47214
11	11	1827.9387	0.16324	0.03708	0.00001	1.71084	0.23476	0	-0.06478	0.00866	0	13.20558	All	55.10094
14	11	1819.81479	0.15492	0.03717	0.00003	1.92657	0.25617	0	-0.07248	0.00937	0	13.2904	All	46.97702
17	11	1810.08794	0.15022	0.03743	0.00006	2.24007	0.29105	0	-0.08307	0.01049	0	13.48303	All	37.25017
20	11	1805.3947	0.14855	0.03777	0.00008	2.52927	0.32331	0	-0.09233	0.01149	0	13.69761	All	32.55694
23	11	1811.41303	0.14978	0.03845	0.0001	2.58133	0.32928	0	-0.09336	0.01161	0	13.82521	All	38.57527
26	11	1819.80331	0.15272	0.03893	0.00009	2.56828	0.32823	0	-0.09201	0.01148	0	13.95702	All	46.96554
29	11	1827.10753	0.1576	0.03941	0.00006	2.58149	0.33032	0	-0.09151	0.01145	0	14.10447	All	54.26976
32	11	1832.24239	0.16209	0.03989	0.00005	2.61491	0.33393	0	-0.09205	0.0115	0	14.20306	All	59.40463
35	11	1835.09705	0.16448	0.03986	0.00004	2.69153	0.34228	0	-0.09416	0.01172	0	14.29194	All	62.25928
2	14	1880.65521	0.22434	0.04182	0	0.68064	0.11978	0	-0.02646	0.00458	0	12.86198	All	107.81744
5	14	1850.238	0.22315	0.04054	0	1.16065	0.17285	0	-0.04403	0.00646	0	13.17937	All	77.40024
8	14	1830.5969	0.221	0.04019	0	1.5239	0.21437	0	-0.05781	0.00795	0	13.18073	All	57.75913
11	14	1820.85763	0.21404	0.04092	0	1.75356	0.24004	0	-0.06637	0.00884	0	13.21084	All	48.01987
14	14	1813.84242	0.20201	0.04124	0	1.95534	0.26009	0	-0.0735	0.0095	0	13.30142	All	41.00466
17	14	1804.02247	0.19802	0.04143	0	2.26588	0.29469	0	-0.08396	0.01061	0	13.49365	All	31.18471
20	14	1799.36266	0.19622	0.04163	0	2.55943	0.32743	0	-0.09335	0.01162	0	13.70887	All	26.52489
23	14	1805.85284	0.19535	0.04225	0	2.60448	0.33228	0	-0.09414	0.0117	0	13.83355	All	33.01508
26	14	1814.50865	0.19738	0.04277	0	2.58144	0.33015	0	-0.09245	0.01153	0	13.96173	All	41.67089
29	14	1821.99319	0.20152	0.04328	0	2.59298	0.33245	0	-0.09189	0.01151	0	14.10925	All	49.15542
32	14	1827.59728	0.20389	0.04381	0	2.6208	0.33604	0	-0.09224	0.01156	0	14.20652	All	54.75951
35	14	1830.77049	0.20546	0.04403	0	2.68926	0.34385	0	-0.09407	0.01176	0	14.29337	All	57.93273
2	17	1873.8011	0.27289	0.04487	0	0.69178	0.12098	0	-0.02701	0.00462	0	12.80728	All	100.96334
5	17	1843.62897	0.26972	0.04369	0	1.17357	0.17384	0	-0.04462	0.00649	0	13.1498	All	70.79121
8	17	1823.83039	0.26667	0.04321	0	1.54223	0.21642	0	-0.05855	0.00801	0	13.17067	All	50.99263
11	17	1813.31567	0.26363	0.04387	0	1.79987	0.24596	0	-0.06809	0.00905	0	13.2167	All	40.47791
14	17	1806.65856	0.25137	0.0442	0	2.01425	0.26699	0	-0.07564	0.00974	0	13.3153	All	33.8208
17	17	1797.78107	0.24358	0.04446	0	2.31925	0.30153	0	-0.08585	0.01085	0	13.50793	All	24.94331
20	17	1793.32796	0.24024	0.0445	0	2.60604	0.33322	0	-0.09497	0.01182	0	13.72061	All	20.49019
23	17	1800.23281	0.23729	0.04502	0	2.64239	0.337	0	-0.09544	0.01186	0	13.84386	All	27.39505
26	17	1809.43955	0.23679	0.04551	0	2.61179	0.33428	0	-0.09347	0.01167	0	13.97159	All	36.60179
29	17	1817.0778	0.2411	0.04617	0	2.61528	0.33609	0	-0.09262	0.01162	0	14.11876	All	44.24003
32	17	1822.70324	0.24448	0.04687	0	2.64197	0.33998	0	-0.09291	0.01168	0	14.21763	All	49.86548
35	17	1826.37033	0.24376	0.04708	0	2.7041	0.34792	0	-0.09452	0.01188	0	14.30409	All	53.53257
2	20	1867.50518	0.33212	0.05064	0	0.68771	0.12085	0	-0.02706	0.00462	0	12.7085	All	94.66742
5	20	1837.10699	0.32706	0.04899	0	1.17776	0.17471	0	-0.04497	0.00652	0	13.09455	All	64.26922
8	20	1817.19105	0.32226	0.04822	0	1.55388	0.21791	0	-0.05911	0.00806	0	13.14333	All	44.35329

11	20	1806.18061	0.32108	0.04888	0	1.82828	0.24936	0	-0.06921	0.00916	0	13.20775	All	33.34284
14	20	1798.86899	0.31176	0.04916	0	2.07231	0.27393	0	-0.07779	0.00998	0	13.31989	All	26.03123
17	20	1789.82211	0.30559	0.04946	0	2.40374	0.31112	0	-0.08892	0.01119	0	13.51576	All	16.98434
20	20	1786.54851	0.29651	0.04945	0	2.6773	0.34165	0	-0.09752	0.01212	0	13.72686	All	13.71074
23	20	1794.04328	0.29032	0.04981	0	2.6905	0.34321	0	-0.09714	0.01207	0	13.84823	All	21.20552
26	20	1803.67182	0.28759	0.05024	0	2.64807	0.33977	0	-0.09475	0.01185	0	13.97454	All	30.83406
29	20	1811.75176	0.29009	0.05092	0	2.64515	0.34159	0	-0.09366	0.0118	0	14.12178	All	38.914
32	20	1817.57836	0.29356	0.05177	0	2.66599	0.34545	0	-0.09373	0.01185	0	14.22179	All	44.7406
35	20	1821.33759	0.29329	0.05212	0	2.72721	0.35385	0	-0.09529	0.01207	0	14.30974	All	48.49983
2	23	1860.75953	0.39998	0.05761	0	0.69224	0.12114	0	-0.02742	0.00461	0	12.62218	All	87.92176
5	23	1830.95947	0.38973	0.0556	0	1.17718	0.1744	0	-0.04515	0.0065	0	13.03737	All	58.1217
8	23	1811.14859	0.38068	0.05421	0	1.55474	0.21841	0	-0.05935	0.00808	0	13.09837	All	38.31083
11	23	1799.97129	0.37838	0.05457	0	1.84013	0.2511	0	-0.06983	0.00923	0	13.17628	All	27.13352
14	23	1791.97675	0.37191	0.05491	0	2.10456	0.27774	0	-0.07911	0.01012	0	13.3008	All	19.13899
17	23	1781.71217	0.37177	0.0553	0	2.48319	0.3197	0	-0.09193	0.0115	0	13.50602	All	8.87441
20	23	1778.56611	0.36177	0.05511	0	2.77868	0.35169	0	-0.10127	0.01249	0	13.71886	All	5.72835
23	23	1787.44024	0.34847	0.05525	0	2.76855	0.35161	0	-0.1	0.01237	0	13.84296	All	14.60247
26	23	1797.74428	0.3413	0.05538	0	2.70429	0.34628	0	-0.0968	0.01207	0	13.96901	All	24.90651
29	23	1806.23103	0.34137	0.05593	0	2.69393	0.34791	0	-0.09543	0.01201	0	14.11537	All	33.39327
32	23	1812.42263	0.34361	0.05681	0	2.71243	0.35222	0	-0.0954	0.01208	0	14.21589	All	39.58486
35	23	1816.24603	0.34475	0.05748	0	2.77348	0.36112	0	-0.09693	0.01231	0	14.306	All	43.40827
2	26	1858.45171	0.4512	0.06483	0	0.69228	0.12136	0	-0.02752	0.00461	0	12.57865	All	85.61395
5	26	1827.6633	0.44864	0.06327	0	1.17934	0.17398	0	-0.04531	0.00646	0	13.01446	All	54.82554
8	26	1807.87944	0.43669	0.06152	0	1.55437	0.21796	0	-0.05945	0.00805	0	13.07243	All	35.04168
11	26	1796.50261	0.43185	0.06127	0	1.84514	0.25174	0	-0.0702	0.00925	0	13.14259	All	23.66484
14	26	1788.09271	0.42556	0.06139	0	2.12174	0.27959	0	-0.07994	0.0102	0	13.27136	All	15.25494
17	26	1777.3128	0.4277	0.06184	0	2.52044	0.32374	0	-0.09345	0.01166	0	13.48581	All	4.47503
20	26	1772.83776	0.42404	0.06175	0	2.86579	0.36005	0	-0.10455	0.0128	0	13.70482	All	0
23	26	1781.97449	0.40804	0.06157	0	2.8646	0.36031	0	-0.10356	0.01269	0	13.83077	All	9.13673
26	26	1793.5908	0.39279	0.06138	0	2.77725	0.3534	0	-0.09948	0.01233	0	13.95892	All	20.75303
29	26	1802.60266	0.38861	0.06159	0	2.75191	0.35377	0	-0.09755	0.01222	0	14.10531	All	29.76489
32	26	1809.14291	0.38876	0.0624	0	2.76573	0.358	0	-0.09734	0.01228	0	14.20629	All	36.30514
35	26	1813.2462	0.38891	0.06316	0	2.83099	0.36803	0	-0.09899	0.01254	0	14.2993	All	40.40844
2	29	1864.62421	0.46241	0.07237	0	0.67226	0.11961	0	-0.02684	0.00456	0	12.52477	All	91.78644
5	29	1833.96506	0.4648	0.0713	0	1.157	0.17208	0	-0.04454	0.0064	0	12.9873	All	61.1273
8	29	1812.60645	0.4647	0.06991	0	1.53815	0.21562	0	-0.05891	0.00796	0	13.05469	All	39.76868
11	29	1800.46675	0.46134	0.06917	0	1.8383	0.25034	0	-0.07002	0.00919	0	13.12643	All	27.62899
14	29	1791.56759	0.45393	0.06869	0	2.12122	0.27921	0	-0.08007	0.01019	0	13.24525	All	18.72983

17	29	1780.75213	0.45493	0.06884	0	2.52189	0.3237	0	-0.09367	0.01166	0	13.46183	All	7.91436
20	29	1775.72125	0.45391	0.06885	0	2.88665	0.36269	0	-0.10544	0.0129	0	13.6882	All	2.88349
23	29	1783.5931	0.4424	0.0687	0	2.92347	0.36622	0	-0.10579	0.01291	0	13.81715	All	10.75534
26	29	1795.27794	0.42377	0.06822	0	2.84364	0.3594	0	-0.10195	0.01255	0	13.94587	All	22.44017
29	29	1805.07954	0.41269	0.0681	0	2.80692	0.35893	0	-0.09957	0.01241	0	14.09495	All	32.24178
32	29	1811.69489	0.41022	0.06866	0	2.8158	0.36252	0	-0.09916	0.01244	0	14.19835	All	38.85712
35	29	1815.91224	0.40872	0.06942	0	2.88174	0.37264	0	-0.1008	0.0127	0	14.29401	All	43.07448
2	32	1868.13968	0.47679	0.07897	0	0.67185	0.11908	0	-0.02684	0.00456	0	12.5143	All	95.30192
5	32	1839.16469	0.47169	0.07811	0	1.13305	0.16937	0	-0.04371	0.00632	0	12.96181	All	66.32693
8	32	1817.71706	0.47964	0.07791	0	1.51313	0.21249	0	-0.05808	0.00787	0	13.02618	All	44.8793
11	32	1804.3195	0.48727	0.07775	0	1.82567	0.24839	0	-0.06965	0.00914	0	13.10568	All	31.48174
14	32	1794.37192	0.48507	0.07707	0	2.13147	0.28016	0	-0.08055	0.01023	0	13.23096	All	21.53416
17	32	1783.16815	0.48658	0.07677	0	2.54138	0.32566	0	-0.09453	0.01175	0	13.44285	All	10.33039
20	32	1778.0408	0.48504	0.07655	0	2.91234	0.36528	0	-0.10652	0.01301	0	13.67058	All	5.20304
23	32	1785.31438	0.4752	0.07635	0	2.96723	0.37078	0	-0.10748	0.01309	0	13.80399	All	12.47661
26	32	1796.23582	0.4584	0.07569	0	2.90645	0.36526	0	-0.10431	0.01278	0	13.93233	All	23.39806
29	32	1806.25008	0.44358	0.07526	0	2.86906	0.36424	0	-0.10187	0.01262	0	14.08146	All	33.41232
32	32	1813.39745	0.43536	0.07547	0	2.86972	0.36708	0	-0.10112	0.01262	0	14.18912	All	40.55969
35	32	1817.6061	0.43172	0.07592	0	2.93419	0.37693	0	-0.10268	0.01286	0	14.28808	All	44.76833
2	35	1870.31492	0.49326	0.08517	0	0.68041	0.11963	0	-0.02718	0.00458	0	12.51732	All	97.47715
5	35	1840.92217	0.49531	0.08526	0	1.14142	0.16879	0	-0.04411	0.00631	0	12.93853	All	68.08441
8	35	1820.08923	0.50294	0.08572	0	1.5091	0.20972	0	-0.05807	0.00779	0	12.99419	All	47.25147
11	35	1806.79788	0.51583	0.08667	0	1.82195	0.2454	0	-0.06967	0.00906	0	13.07539	All	33.96011
14	35	1796.08105	0.52253	0.08672	0	2.13723	0.27877	0	-0.08091	0.01021	0	13.20821	All	23.24328
17	35	1784.21205	0.52942	0.08649	0	2.5662	0.32622	0	-0.09557	0.01179	0	13.4255	All	11.37429
20	35	1779.04586	0.52703	0.08597	0	2.94386	0.36662	0	-0.10783	0.01308	0	13.65001	All	6.2081
23	35	1786.56389	0.51237	0.08524	0	2.99981	0.37254	0	-0.1088	0.01317	0	13.78535	All	13.72613
26	35	1797.00993	0.49583	0.08423	0	2.95581	0.36877	0	-0.10619	0.01292	0	13.91722	All	24.17217
29	35	1806.64262	0.4807	0.08337	0	2.92951	0.36858	0	-0.10416	0.01279	0	14.06285	All	33.80486
32	35	1814.06744	0.46781	0.08306	0	2.92821	0.37104	0	-0.10333	0.01278	0	14.16875	All	41.22968
35	35	1818.62088	0.4593	0.08304	0	2.99092	0.38077	0	-0.10478	0.01302	0	14.2724	All	45.78312
2	2	858.33355	0.01164	0.0339	0.73124	0.47007	0.15609	0.0026	-0.01989	0.00622	0.00137	11.81448	Site_A	58.62309
5	2	839.81624	0.01691	0.03349	0.61354	1.06242	0.25549	0.00003	-0.04276	0.00993	0.00002	12.42263	Site_A	40.10578
8	2	833.79635	0.01936	0.03353	0.56363	1.27323	0.29262	0.00001	-0.05147	0.01136	0.00001	12.36954	Site_A	34.08589
11	2	824.2589	0.01564	0.03397	0.64523	1.67178	0.36555	0	-0.0665	0.014	0	12.56913	Site_A	24.54844
14	2	818.29706	0.01126	0.03397	0.74033	2.01023	0.42441	0	-0.07884	0.01607	0	12.74938	Site_A	18.5866
17	2	816.32044	0.008	0.03394	0.81362	2.18892	0.45654	0	-0.08493	0.01707	0	12.88613	Site_A	16.60997
20	2	815.56394	0.00479	0.03414	0.88853	2.3759	0.49318	0	-0.09077	0.01814	0	13.08803	Site_A	15.85348

23	2	816.6322	0.00387	0.03425	0.90996	2.51037	0.51668	0	-0.09446	0.01874	0	13.28826	Site_A	16.92174
26	2	819.87871	0.00505	0.03442	0.88334	2.5344	0.51895	0	-0.09416	0.01859	0	13.45777	Site_A	20.16825
29	2	823.28152	0.007	0.03448	0.8391	2.55875	0.52117	0	-0.09393	0.01847	0	13.62104	Site_A	23.57106
32	2	824.92645	0.00767	0.03441	0.82366	2.6545	0.53372	0	-0.09648	0.01875	0	13.75739	Site_A	25.21599
35	2	827.10623	0.0086	0.03431	0.80201	2.70585	0.5415	0	-0.09755	0.01887	0	13.86862	Site_A	27.39577
2	5	856.45105	0.06584	0.04457	0.13955	0.45223	0.15536	0.0036	-0.01909	0.00619	0.00204	11.84735	Site_A	56.74059
5	5	838.2525	0.06244	0.04451	0.16065	1.03962	0.25616	0.00005	-0.04182	0.00995	0.00003	12.4292	Site_A	38.54204
8	5	831.76364	0.07162	0.04455	0.10787	1.25037	0.29231	0.00002	-0.05061	0.01135	0.00001	12.35337	Site_A	32.05318
11	5	822.22069	0.07077	0.04527	0.11798	1.64519	0.3648	0.00001	-0.06557	0.01398	0	12.54582	Site_A	22.51023
14	5	816.61527	0.06316	0.04544	0.16454	1.97346	0.42268	0	-0.07752	0.01602	0	12.72929	Site_A	16.90481
17	5	814.85143	0.05849	0.04572	0.20085	2.14747	0.45525	0	-0.08344	0.01704	0	12.86875	Site_A	15.14097
20	5	814.28517	0.05421	0.04607	0.23929	2.32551	0.49145	0	-0.08896	0.01809	0	13.07088	Site_A	14.57471
23	5	815.46732	0.05217	0.04667	0.26362	2.44797	0.51379	0	-0.09225	0.01865	0	13.26871	Site_A	15.75686
26	5	818.63276	0.05459	0.04701	0.24548	2.46108	0.51407	0	-0.09159	0.01843	0	13.43509	Site_A	18.9223
29	5	821.8911	0.05853	0.04736	0.2165	2.4787	0.51515	0	-0.09114	0.01826	0	13.59877	Site_A	22.18064
32	5	823.43399	0.06076	0.04734	0.19931	2.57097	0.52683	0	-0.09359	0.01852	0	13.73496	Site_A	23.72353
35	5	825.55714	0.06193	0.04717	0.18916	2.62038	0.53426	0	-0.09466	0.01864	0	13.84095	Site_A	25.84668
2	8	857.7981	0.04776	0.05775	0.40825	0.46419	0.1561	0.00294	-0.01957	0.00622	0.00165	11.86276	Site_A	58.08764
5	8	839.64266	0.03803	0.05768	0.5097	1.05856	0.25693	0.00004	-0.04251	0.00998	0.00002	12.45023	Site_A	39.9322
8	8	833.70992	0.03755	0.05802	0.51748	1.26539	0.29371	0.00002	-0.05108	0.01139	0.00001	12.38733	Site_A	33.99946
11	8	824.07565	0.03706	0.05853	0.52662	1.66232	0.3663	0.00001	-0.06609	0.01403	0	12.57713	Site_A	24.36519
14	8	818.16902	0.02887	0.05878	0.62337	1.99829	0.42487	0	-0.07835	0.01609	0	12.7519	Site_A	18.45856
17	8	816.23939	0.02213	0.05951	0.70998	2.17728	0.4574	0	-0.08448	0.01711	0	12.88694	Site_A	16.52893
20	8	815.5048	0.01696	0.06003	0.77756	2.36512	0.49434	0	-0.09035	0.01819	0	13.08836	Site_A	15.79434
23	8	816.60221	0.01263	0.06077	0.83542	2.50091	0.51871	0	-0.0941	0.01882	0	13.28847	Site_A	16.89175
26	8	819.85639	0.0129	0.06143	0.83368	2.52445	0.5222	0	-0.09379	0.01872	0	13.45794	Site_A	20.14593
29	8	823.25579	0.01607	0.06199	0.79549	2.54477	0.52557	0	-0.09342	0.01863	0	13.62059	Site_A	23.54533
32	8	824.88431	0.01898	0.06246	0.76122	2.63552	0.5384	0	-0.09579	0.01892	0	13.75669	Site_A	25.17385
35	8	827.04468	0.02198	0.06207	0.72321	2.68251	0.54589	0	-0.09672	0.01903	0	13.86738	Site_A	27.33422
2	11	856.04022	0.10321	0.06408	0.10724	0.45754	0.15617	0.00339	-0.01925	0.00621	0.00194	11.8871	Site_A	56.32976
5	11	838.07657	0.09247	0.06329	0.144	1.05458	0.25834	0.00004	-0.04224	0.01002	0.00002	12.4839	Site_A	38.3661
8	11	832.35256	0.08773	0.06391	0.16985	1.2619	0.29562	0.00002	-0.05082	0.01145	0.00001	12.41504	Site_A	32.6421
11	11	822.96834	0.08127	0.06457	0.20817	1.65395	0.36788	0.00001	-0.06566	0.01407	0	12.59404	Site_A	23.25788
14	11	817.18491	0.07335	0.06478	0.25751	1.97955	0.42496	0	-0.07757	0.01608	0	12.75958	Site_A	17.47445
17	11	815.36923	0.06707	0.06538	0.30496	2.15012	0.45647	0	-0.08341	0.01706	0	12.88957	Site_A	15.65877
20	11	814.73652	0.06195	0.06592	0.34733	2.33106	0.49269	0	-0.08905	0.01813	0	13.08823	Site_A	15.02606
23	11	815.89312	0.05904	0.06676	0.37651	2.45765	0.51577	0	-0.0925	0.01872	0	13.28522	Site_A	16.18266
26	11	819.15288	0.05947	0.06748	0.37814	2.47268	0.51766	0	-0.0919	0.01856	0	13.45298	Site_A	19.44242

29	11	822.53669	0.06162	0.06818	0.36607	2.48741	0.52013	0	-0.09135	0.01844	0	13.61473	Site_A	22.82623
32	11	824.15144	0.06386	0.06897	0.35449	2.57284	0.53244	0	-0.09356	0.01871	0	13.7491	Site_A	24.44098
35	11	826.27438	0.06652	0.06891	0.33443	2.61617	0.54	0	-0.0944	0.01883	0	13.85751	Site_A	26.56392
2	14	853.23679	0.1661	0.06995	0.01757	0.45353	0.15663	0.00379	-0.01912	0.00622	0.00212	11.8594	Site_A	53.52633
5	14	835.35172	0.15478	0.06853	0.0239	1.05634	0.26009	0.00005	-0.04228	0.01006	0.00003	12.49096	Site_A	35.64126
8	14	829.74978	0.14981	0.06892	0.02973	1.26655	0.29866	0.00002	-0.05093	0.01154	0.00001	12.43501	Site_A	30.03932
11	14	820.59736	0.14225	0.06975	0.04141	1.66401	0.37321	0.00001	-0.06594	0.01424	0	12.61839	Site_A	20.8869
14	14	815.18396	0.13031	0.07034	0.06394	1.97698	0.42856	0	-0.07736	0.01618	0	12.77806	Site_A	15.4735
17	14	813.45459	0.12483	0.07091	0.07834	2.13594	0.45848	0	-0.08279	0.01711	0	12.89997	Site_A	13.74413
20	14	812.91933	0.11963	0.07121	0.09296	2.30979	0.49394	0	-0.0882	0.01815	0	13.0946	Site_A	13.20887
23	14	814.15599	0.11663	0.07188	0.10467	2.42867	0.51551	0	-0.09139	0.01869	0	13.28768	Site_A	14.44553
26	14	817.4309	0.11726	0.07256	0.10607	2.43416	0.51542	0	-0.09047	0.01847	0	13.45221	Site_A	17.72044
29	14	820.82011	0.11916	0.07324	0.10374	2.44223	0.51649	0	-0.0897	0.0183	0	13.61333	Site_A	21.10965
32	14	822.50449	0.11987	0.07416	0.106	2.52066	0.52777	0	-0.09169	0.01853	0	13.74567	Site_A	22.79403
35	14	824.68668	0.12066	0.07447	0.10518	2.55688	0.53475	0	-0.09231	0.01864	0	13.85003	Site_A	24.97622
2	17	852.19784	0.19857	0.07632	0.00927	0.45195	0.15684	0.00396	-0.01915	0.00622	0.00209	11.80193	Site_A	52.48738
5	17	834.48981	0.1841	0.07493	0.014	1.05654	0.26088	0.00005	-0.04233	0.01008	0.00003	12.4787	Site_A	34.77935
8	17	828.91628	0.178	0.0751	0.01778	1.26913	0.30038	0.00002	-0.05103	0.01159	0.00001	12.43547	Site_A	29.20581
11	17	819.71204	0.17171	0.07589	0.02367	1.68072	0.3781	0.00001	-0.0665	0.01439	0	12.63691	Site_A	20.00158
14	17	814.36896	0.1593	0.07663	0.03764	2.00509	0.43504	0	-0.0783	0.0164	0	12.80344	Site_A	14.6585
17	17	812.86573	0.14943	0.07734	0.05333	2.15704	0.4638	0	-0.08347	0.01729	0	12.92171	Site_A	13.15526
20	17	812.35987	0.14318	0.0774	0.06435	2.32557	0.49816	0	-0.08869	0.01829	0	13.11096	Site_A	12.64941
23	17	813.62387	0.1393	0.07785	0.07358	2.44151	0.51893	0	-0.09178	0.0188	0	13.30114	Site_A	13.91341
26	17	816.96759	0.13835	0.07851	0.07805	2.44328	0.51785	0	-0.09074	0.01854	0	13.46383	Site_A	17.25713
29	17	820.39579	0.13981	0.07942	0.07837	2.44739	0.51833	0	-0.08982	0.01835	0	13.62416	Site_A	20.68532
32	17	822.11258	0.14019	0.08055	0.0818	2.52412	0.52962	0	-0.09174	0.01858	0	13.75716	Site_A	22.40212
35	17	824.41081	0.13821	0.08094	0.08772	2.55731	0.53689	0	-0.09225	0.0187	0	13.86093	Site_A	24.70035
2	20	846.9174	0.29349	0.08379	0.00046	0.43676	0.15691	0.00538	-0.01874	0.00623	0.00262	11.65117	Site_A	47.20694
5	20	829.58716	0.27373	0.08169	0.00081	1.04367	0.26246	0.00007	-0.04198	0.01013	0.00003	12.42951	Site_A	29.8767
8	20	824.10913	0.26683	0.08149	0.00106	1.26227	0.30409	0.00003	-0.05084	0.01171	0.00001	12.41333	Site_A	24.39866
11	20	814.99601	0.26073	0.08189	0.00145	1.69018	0.38614	0.00001	-0.06685	0.01467	0.00001	12.64095	Site_A	15.28555
14	20	809.71755	0.25157	0.08256	0.00231	2.03569	0.44815	0.00001	-0.07935	0.01684	0	12.82659	Site_A	10.00709
17	20	808.38522	0.24311	0.08334	0.00353	2.19222	0.4785	0	-0.08465	0.01779	0	12.94825	Site_A	8.67476
20	20	808.24357	0.23263	0.08334	0.00525	2.3436	0.51065	0	-0.08925	0.01872	0	13.1299	Site_A	8.53311
23	20	809.68139	0.22686	0.0835	0.00659	2.44062	0.52822	0	-0.09167	0.01911	0	13.31138	Site_A	9.97093
26	20	813.10783	0.22532	0.08402	0.00733	2.42653	0.52438	0	-0.09008	0.01874	0	13.46882	Site_A	13.39737
29	20	816.58705	0.22665	0.08486	0.00757	2.41871	0.52269	0	-0.08874	0.01847	0	13.62737	Site_A	16.87659
32	20	818.31331	0.22867	0.08607	0.00789	2.48623	0.5326	0	-0.09034	0.01865	0	13.75973	Site_A	18.60285

35	20	820.67802	0.22707	0.08659	0.00873	2.50796	0.53856	0	-0.09047	0.01872	0	13.86011	Site_A	20.96756
2	23	844.78951	0.35512	0.09457	0.00017	0.43394	0.15676	0.00564	-0.01878	0.00621	0.00248	11.55272	Site_A	45.07905
5	23	827.77383	0.32981	0.0922	0.00035	1.03499	0.26179	0.00008	-0.04179	0.0101	0.00003	12.38249	Site_A	28.06337
8	23	822.41377	0.31938	0.09139	0.00047	1.25108	0.30386	0.00004	-0.05056	0.0117	0.00002	12.37144	Site_A	22.70331
11	23	813.36099	0.31061	0.0911	0.00065	1.68757	0.38781	0.00001	-0.06686	0.01473	0.00001	12.61986	Site_A	13.65053
14	23	807.95797	0.30281	0.09159	0.00095	2.04979	0.45323	0.00001	-0.07994	0.01703	0	12.82129	Site_A	8.24751
17	23	806.37579	0.29918	0.09258	0.00123	2.23512	0.48928	0	-0.08624	0.01817	0	12.95856	Site_A	6.66533
20	23	806.26465	0.28827	0.09241	0.00181	2.40222	0.52377	0	-0.09136	0.01918	0	13.14683	Site_A	6.55419
23	23	807.96091	0.27787	0.09234	0.00262	2.48817	0.53912	0	-0.09336	0.01949	0	13.32618	Site_A	8.25045
26	23	811.54688	0.2728	0.09246	0.00317	2.4579	0.5323	0	-0.09118	0.01901	0	13.47884	Site_A	11.83642
29	23	815.10207	0.27257	0.09313	0.00342	2.44347	0.52975	0	-0.0896	0.0187	0	13.63539	Site_A	15.39161
32	23	816.90499	0.27369	0.09434	0.00372	2.50899	0.53954	0	-0.09112	0.01888	0	13.76742	Site_A	17.19453
35	23	819.30592	0.27261	0.09521	0.00419	2.52578	0.54501	0	-0.09106	0.01893	0	13.86805	Site_A	19.59546
2	26	839.82954	0.45503	0.10644	0.00002	0.42385	0.15711	0.00698	-0.01855	0.0062	0.00276	11.42601	Site_A	40.11908
5	26	822.97386	0.4284	0.10393	0.00004	1.02078	0.26118	0.00009	-0.04133	0.01003	0.00004	12.34995	Site_A	23.2634
8	26	817.74353	0.41559	0.10304	0.00005	1.2351	0.30422	0.00005	-0.05009	0.01168	0.00002	12.32982	Site_A	18.03307
11	26	808.75686	0.40442	0.10208	0.00007	1.68183	0.3914	0.00002	-0.0668	0.01485	0.00001	12.58801	Site_A	9.0464
14	26	803.29386	0.3975	0.10222	0.0001	2.06517	0.46195	0.00001	-0.08064	0.01735	0	12.80442	Site_A	3.5834
17	26	801.63096	0.39655	0.10336	0.00012	2.26961	0.50274	0.00001	-0.08759	0.01866	0	12.9554	Site_A	1.9205
20	26	801.33159	0.38952	0.10317	0.00016	2.46995	0.54321	0.00001	-0.09385	0.01988	0	13.15839	Site_A	1.62113
23	26	803.08475	0.37865	0.10267	0.00023	2.56749	0.55934	0	-0.09621	0.02021	0	13.34257	Site_A	3.37429
26	26	807.06476	0.36745	0.10235	0.00033	2.50814	0.54737	0	-0.09296	0.01954	0	13.49041	Site_A	7.3543
29	26	810.81403	0.3637	0.10253	0.00039	2.47041	0.54106	0	-0.09055	0.01908	0	13.64118	Site_A	11.10357
32	26	812.70151	0.36417	0.10355	0.00044	2.52891	0.55017	0	-0.09182	0.01922	0	13.77151	Site_A	12.99105
35	26	815.21934	0.36254	0.10448	0.00052	2.53962	0.5554	0	-0.09154	0.01926	0	13.87185	Site_A	15.50888
2	29	840.16357	0.49816	0.11971	0.00003	0.41651	0.15583	0.00752	-0.01839	0.00617	0.00287	11.32696	Site_A	40.45311
5	29	823.49778	0.46872	0.11734	0.00006	1.00743	0.25895	0.0001	-0.0409	0.00995	0.00004	12.31665	Site_A	23.78732
8	29	817.89568	0.46139	0.11665	0.00008	1.22548	0.30219	0.00005	-0.04975	0.01159	0.00002	12.31595	Site_A	18.18522
11	29	808.7458	0.4506	0.11537	0.00009	1.67901	0.3904	0.00002	-0.06674	0.0148	0.00001	12.57891	Site_A	9.03534
14	29	803.14681	0.4432	0.11494	0.00012	2.06967	0.46295	0.00001	-0.08091	0.01738	0	12.79	Site_A	3.43635
17	29	801.4572	0.44172	0.11587	0.00014	2.28157	0.50569	0.00001	-0.08812	0.01877	0	12.94509	Site_A	1.74674
20	29	801.03059	0.43632	0.11582	0.00016	2.50103	0.55044	0.00001	-0.09503	0.02013	0	13.15983	Site_A	1.32013
23	29	802.43381	0.42969	0.11522	0.00019	2.63486	0.57195	0	-0.09865	0.02066	0	13.35392	Site_A	2.72335
26	29	806.38201	0.41694	0.11438	0.00027	2.58781	0.56027	0	-0.0958	0.02	0	13.50577	Site_A	6.67155
29	29	810.38531	0.40739	0.11406	0.00035	2.53724	0.55172	0	-0.09289	0.01946	0	13.6565	Site_A	10.67484
32	29	812.37614	0.40458	0.11465	0.00042	2.58612	0.55905	0	-0.0938	0.01953	0	13.78539	Site_A	12.66568
35	29	814.9726	0.40102	0.11555	0.00052	2.59373	0.56383	0	-0.09339	0.01954	0	13.88687	Site_A	15.26214
2	32	839.03748	0.56034	0.13337	0.00003	0.42265	0.15556	0.00659	-0.01869	0.00619	0.00253	11.30567	Site_A	39.32702

5	32	822.77943	0.52422	0.13122	0.00006	0.99558	0.25626	0.0001	-0.0406	0.00988	0.00004	12.26199	Site_A	23.06897
8	32	817.15777	0.52075	0.13158	0.00008	1.21703	0.30085	0.00005	-0.04955	0.01156	0.00002	12.27994	Site_A	17.44731
11	32	807.71726	0.51587	0.13078	0.00008	1.68208	0.39157	0.00002	-0.06695	0.01486	0.00001	12.56256	Site_A	8.0068
14	32	801.86077	0.51212	0.13036	0.00009	2.09465	0.46887	0.00001	-0.08193	0.01761	0	12.78291	Site_A	2.15031
17	32	800.1135	0.50973	0.13071	0.0001	2.31604	0.51404	0.00001	-0.08952	0.01909	0	12.93592	Site_A	0.40303
20	32	799.71046	0.50322	0.13049	0.00012	2.54588	0.56124	0.00001	-0.09677	0.02054	0	13.15458	Site_A	0
23	32	800.98971	0.49836	0.12994	0.00013	2.69871	0.58619	0	-0.10103	0.02118	0	13.35592	Site_A	1.27925
26	32	804.70995	0.48776	0.12873	0.00015	2.66903	0.57577	0	-0.09877	0.02057	0	13.51106	Site_A	4.99949
29	32	808.73743	0.47608	0.12784	0.0002	2.6203	0.5659	0	-0.09588	0.01997	0	13.6642	Site_A	9.02697
32	32	810.93052	0.46826	0.1279	0.00025	2.65979	0.57169	0	-0.0964	0.01998	0	13.79554	Site_A	11.22006
35	32	813.64158	0.46104	0.12826	0.00032	2.6557	0.57461	0	-0.09554	0.01992	0	13.89772	Site_A	13.93112
2	35	842.28199	0.54667	0.14259	0.00013	0.44094	0.15628	0.00478	-0.0193	0.00622	0.00192	11.42369	Site_A	42.57153
5	35	825.23455	0.52489	0.14232	0.00023	1.02281	0.25601	0.00006	-0.04166	0.00991	0.00003	12.27663	Site_A	25.52409
8	35	819.76059	0.51901	0.14304	0.00029	1.23977	0.29886	0.00003	-0.05049	0.01154	0.00001	12.2785	Site_A	20.05013
11	35	810.3523	0.51757	0.14382	0.00032	1.70078	0.38835	0.00001	-0.06773	0.01479	0	12.55515	Site_A	10.64184
14	35	804.25175	0.52107	0.1446	0.00031	2.11652	0.46676	0.00001	-0.0828	0.01757	0	12.78104	Site_A	4.54129
17	35	802.34839	0.52053	0.14497	0.00033	2.34926	0.51393	0	-0.09078	0.01911	0	12.9391	Site_A	2.63793
20	35	801.94298	0.512	0.14435	0.00039	2.5798	0.56125	0	-0.09807	0.02056	0	13.15247	Site_A	2.23252
23	35	803.32438	0.50389	0.14352	0.00045	2.73239	0.58641	0	-0.10231	0.02121	0	13.35307	Site_A	3.61392
26	35	806.93166	0.49352	0.14199	0.00051	2.71889	0.57885	0	-0.1006	0.02069	0	13.51349	Site_A	7.2212
29	35	810.75138	0.48308	0.1406	0.00059	2.68981	0.57092	0	-0.09841	0.02018	0	13.66681	Site_A	11.04092
32	35	812.91563	0.47276	0.14002	0.00073	2.74034	0.577	0	-0.0993	0.0202	0	13.79828	Site_A	13.20516
35	35	815.7559	0.45908	0.13944	0.00099	2.73623	0.57939	0	-0.09839	0.02011	0	13.9054	Site_A	16.04544
2	2	805.6533	0.0317	0.02685	0.23777	0.72083	0.21821	0.00096	-0.02779	0.00828	0.00079	12.96905	Site_B	53.81078
5	2	783.7122	0.03931	0.02702	0.14576	1.46788	0.30456	0	-0.0553	0.01122	0	13.27185	Site_B	31.86969
8	2	778.1719	0.04057	0.02707	0.13395	1.73671	0.34699	0	-0.06591	0.01278	0	13.17462	Site_B	26.32938
11	2	768.36085	0.03784	0.02757	0.16993	2.16674	0.4116	0	-0.08212	0.01508	0	13.19172	Site_B	16.51833
14	2	764.13923	0.03657	0.02779	0.18815	2.38653	0.44053	0	-0.09044	0.01608	0	13.19374	Site_B	12.29671
17	2	763.79213	0.03659	0.0277	0.18651	2.47633	0.45649	0	-0.09356	0.01653	0	13.23398	Site_B	11.94961
20	2	762.96765	0.03505	0.02786	0.20832	2.67882	0.49457	0	-0.10001	0.01767	0	13.39216	Site_B	11.12513
23	2	763.28818	0.03531	0.02797	0.20682	2.85995	0.52466	0	-0.10544	0.01853	0	13.56167	Site_B	11.44566
26	2	765.40996	0.03673	0.02816	0.19215	2.96948	0.54098	0	-0.10794	0.01888	0	13.7546	Site_B	13.56744
29	2	769.28783	0.03842	0.02834	0.17525	3.00595	0.54888	0	-0.10775	0.01894	0	13.94854	Site_B	17.44531
32	2	770.74121	0.03903	0.02841	0.16954	3.111	0.56381	0	-0.11047	0.01929	0	14.08067	Site_B	18.89869
35	2	773.36372	0.03959	0.02823	0.16079	3.14746	0.57244	0	-0.11089	0.01943	0	14.1921	Site_B	21.5212
2	5	802.48969	0.08022	0.03572	0.02472	0.6991	0.21969	0.00146	-0.02685	0.00835	0.00131	13.01943	Site_B	50.64717
5	5	781.03711	0.08182	0.03541	0.02086	1.45171	0.30731	0	-0.05462	0.01132	0	13.28815	Site_B	29.19459
8	5	774.95014	0.08885	0.03581	0.0131	1.71241	0.34747	0	-0.06501	0.01279	0	13.17131	Site_B	23.10762

11	5	765.14113	0.08731	0.03655	0.01689	2.13487	0.41122	0	-0.08098	0.01506	0	13.18163	Site_B	13.29861
14	5	761.22623	0.08356	0.03676	0.02304	2.34152	0.43831	0	-0.08881	0.01598	0	13.18216	Site_B	9.38371
17	5	760.71283	0.08591	0.03699	0.02021	2.44308	0.45643	0	-0.09235	0.01652	0	13.22695	Site_B	8.87031
20	5	759.76027	0.08637	0.03728	0.02052	2.6587	0.49684	0	-0.0993	0.01775	0	13.3876	Site_B	7.91775
23	5	760.12249	0.08732	0.03797	0.02147	2.83822	0.52618	0	-0.1047	0.01859	0	13.55447	Site_B	8.27997
26	5	762.10577	0.09057	0.03839	0.01832	2.9398	0.53984	0	-0.10694	0.01885	0	13.74507	Site_B	10.26325
29	5	765.79025	0.09481	0.03896	0.01496	2.96567	0.54508	0	-0.10638	0.0188	0	13.93948	Site_B	13.94774
32	5	767.11868	0.09685	0.03917	0.01342	3.06568	0.55842	0	-0.10895	0.01911	0	14.06926	Site_B	15.27617
35	5	769.61875	0.0981	0.0389	0.01167	3.09915	0.56625	0	-0.10932	0.01923	0	14.1753	Site_B	17.77624
2	8	803.37498	0.09038	0.04495	0.04438	0.71644	0.22941	0.00179	-0.02748	0.00872	0.00162	13.03409	Site_B	51.53246
5	8	782.33207	0.08604	0.04448	0.05309	1.47546	0.30955	0	-0.05542	0.0114	0	13.31219	Site_B	30.48955
8	8	777.05534	0.08485	0.04519	0.06041	1.73609	0.35118	0	-0.06574	0.01293	0	13.20485	Site_B	25.21282
11	8	767.15248	0.08328	0.04628	0.07195	2.15653	0.41411	0	-0.08163	0.01516	0	13.20868	Site_B	15.30996
14	8	763.19116	0.07725	0.04642	0.09612	2.35876	0.44032	0	-0.08934	0.01605	0	13.20049	Site_B	11.34864
17	8	762.89349	0.07699	0.04666	0.09892	2.4454	0.45586	0	-0.09235	0.01649	0	13.2399	Site_B	11.05097
20	8	761.88806	0.07829	0.04697	0.09553	2.65867	0.49576	0	-0.0992	0.0177	0	13.40053	Site_B	10.04554
23	8	762.26746	0.07829	0.04754	0.09961	2.84023	0.5262	0	-0.10466	0.01858	0	13.56822	Site_B	10.42494
26	8	764.46201	0.07966	0.04824	0.09866	2.94292	0.54167	0	-0.10695	0.01891	0	13.75777	Site_B	12.61949
29	8	768.2835	0.08363	0.04894	0.08748	2.9662	0.54752	0	-0.10631	0.01889	0	13.95087	Site_B	16.44098
32	8	769.60208	0.08754	0.04958	0.07747	3.06372	0.56108	0	-0.10877	0.01919	0	14.08358	Site_B	17.75956
35	8	772.12636	0.08958	0.04918	0.06854	3.09437	0.56853	0	-0.10901	0.0193	0	14.1937	Site_B	20.28384
2	11	800.45431	0.13669	0.05132	0.00773	0.72071	0.22851	0.00161	-0.02765	0.00869	0.00146	13.03235	Site_B	48.61179
5	11	779.08252	0.1356	0.0485	0.00517	1.49599	0.31268	0	-0.05614	0.01151	0	13.32342	Site_B	27.24
8	11	774.1745	0.13085	0.04926	0.00789	1.7626	0.35638	0	-0.06669	0.01311	0	13.21582	Site_B	22.33198
11	11	764.8251	0.12367	0.05051	0.01434	2.17722	0.41951	0	-0.08234	0.01533	0	13.22079	Site_B	12.98258
14	11	761.07039	0.11636	0.05088	0.02221	2.36234	0.44285	0	-0.08942	0.01612	0	13.20922	Site_B	9.22787
17	11	760.87754	0.1146	0.05101	0.02467	2.44294	0.45734	0	-0.09221	0.01653	0	13.24617	Site_B	9.03502
20	11	759.96222	0.11433	0.05115	0.02539	2.65462	0.49672	0	-0.09901	0.01772	0	13.40532	Site_B	8.1197
23	11	760.31175	0.1151	0.05166	0.02589	2.83389	0.52617	0	-0.10443	0.01858	0	13.56867	Site_B	8.46923
26	11	762.52746	0.11619	0.05223	0.02612	2.93384	0.54142	0	-0.10663	0.01889	0	13.7573	Site_B	10.68494
29	11	766.44775	0.11879	0.053	0.025	2.95179	0.547	0	-0.1058	0.01886	0	13.94999	Site_B	14.60523
32	11	767.78067	0.12263	0.05379	0.02261	3.04374	0.55978	0	-0.10808	0.01914	0	14.08151	Site_B	15.93815
35	11	770.25546	0.12544	0.05369	0.01948	3.06845	0.56638	0	-0.10811	0.01922	0	14.19127	Site_B	18.41295
2	14	801.24657	0.14253	0.05783	0.01373	0.72953	0.23607	0.002	-0.02807	0.00897	0.00175	12.99283	Site_B	49.40405
5	14	779.53552	0.14553	0.05474	0.00784	1.50902	0.31301	0	-0.05669	0.01151	0	13.30915	Site_B	27.693
8	14	774.35682	0.14332	0.05523	0.00946	1.78527	0.3586	0	-0.06755	0.01318	0	13.21448	Site_B	22.5143
11	14	765.02921	0.13461	0.05639	0.01698	2.21208	0.42424	0	-0.08363	0.0155	0	13.22564	Site_B	13.18669
14	14	761.53512	0.12251	0.05689	0.03127	2.39345	0.4471	0	-0.09054	0.01627	0	13.21694	Site_B	9.6926

17	14	761.38516	0.12005	0.0572	0.03583	2.46605	0.45996	0	-0.09304	0.01661	0	13.2527	Site_B	9.54264
20	14	760.52428	0.11846	0.05719	0.03832	2.67485	0.49864	0	-0.09972	0.01778	0	13.41132	Site_B	8.68177
23	14	760.99098	0.11674	0.05747	0.04223	2.85244	0.52765	0	-0.10506	0.01861	0	13.57509	Site_B	9.14846
26	14	763.27267	0.11643	0.05789	0.0443	2.95223	0.54314	0	-0.10726	0.01894	0	13.76218	Site_B	11.43015
29	14	767.29579	0.11712	0.05851	0.04533	2.97376	0.55003	0	-0.10655	0.01896	0	13.95478	Site_B	15.45327
32	14	768.82346	0.11828	0.05945	0.04665	3.06579	0.56373	0	-0.10882	0.01926	0	14.08692	Site_B	16.98094
35	14	771.40679	0.12047	0.05969	0.04355	3.08881	0.57038	0	-0.10878	0.01934	0	14.19803	Site_B	19.56427
2	17	799.95028	0.17078	0.06381	0.00744	0.73404	0.22677	0.00121	-0.02833	0.00861	0.001	12.95612	Site_B	48.10776
5	17	778.09602	0.1743	0.05856	0.00291	1.52093	0.31371	0	-0.0572	0.01153	0	13.294	Site_B	26.2535
8	17	772.82955	0.17275	0.05882	0.00332	1.80199	0.36085	0	-0.06821	0.01325	0	13.20853	Site_B	20.98703
11	17	763.23713	0.16897	0.06024	0.00503	2.24796	0.43047	0	-0.08496	0.01571	0	13.23017	Site_B	11.39462
14	17	759.9291	0.15553	0.0608	0.01053	2.43059	0.45358	0	-0.0919	0.01648	0	13.22478	Site_B	8.08659
17	17	760.12474	0.14806	0.06105	0.0153	2.49354	0.46539	0	-0.09403	0.01679	0	13.25895	Site_B	8.28222
20	17	759.35228	0.14504	0.06096	0.01734	2.69522	0.50267	0	-0.10046	0.01791	0	13.41503	Site_B	7.50976
23	17	759.90468	0.14171	0.06104	0.02025	2.86905	0.53077	0	-0.10565	0.01871	0	13.57745	Site_B	8.06216
26	17	762.31036	0.13914	0.06124	0.02309	2.96593	0.54579	0	-0.10774	0.01902	0	13.76421	Site_B	10.46784
29	17	766.40616	0.13878	0.06183	0.02479	2.98385	0.55262	0	-0.1069	0.01904	0	13.95594	Site_B	14.56364
32	17	767.97726	0.13946	0.06279	0.02634	3.07635	0.56687	0	-0.10918	0.01936	0	14.08869	Site_B	16.13474
35	17	770.68291	0.13989	0.06314	0.02673	3.0972	0.57376	0	-0.10906	0.01944	0	14.20014	Site_B	18.84039
2	20	796.13476	0.23227	0.07364	0.00161	0.72557	0.20481	0.0004	-0.02818	0.00778	0.00029	12.87408	Site_B	44.29224
5	20	774.19159	0.22708	0.06114	0.0002	1.52773	0.31482	0	-0.05763	0.01157	0	13.25559	Site_B	22.34907
8	20	768.84448	0.22584	0.06117	0.00022	1.81523	0.36305	0	-0.06884	0.01334	0	13.18357	Site_B	17.00197
11	20	759.10789	0.2249	0.06244	0.00032	2.27719	0.43589	0	-0.08614	0.0159	0	13.21855	Site_B	7.26537
14	20	755.70728	0.216	0.06325	0.00064	2.47486	0.46222	0	-0.09357	0.01677	0	13.22487	Site_B	3.86476
17	20	756.0711	0.2082	0.06353	0.00105	2.54165	0.47523	0	-0.09583	0.01712	0	13.26179	Site_B	4.22858
20	20	755.69553	0.20017	0.06323	0.00155	2.73421	0.51175	0	-0.1019	0.01822	0	13.4155	Site_B	3.85302
23	20	756.42007	0.1958	0.07184	0.00642	2.89667	0.55305	0	-0.10668	0.01969	0	13.57585	Site_B	4.57755
26	20	758.96619	0.19369	0.07244	0.0075	2.98769	0.56558	0	-0.10854	0.01992	0	13.76339	Site_B	7.12367
29	20	763.18747	0.19364	0.07375	0.00865	2.99975	0.5703	0	-0.10746	0.01986	0	13.95748	Site_B	11.34496
32	20	764.79152	0.19711	0.07521	0.00877	3.09088	0.58413	0	-0.10966	0.02017	0	14.09364	Site_B	12.94901
35	20	767.50895	0.19936	0.07547	0.00825	3.11222	0.59073	0	-0.10953	0.02025	0	14.20706	Site_B	15.66643
2	23	792.78361	0.29737	0.08614	0.00056	0.7222	0.19576	0.00022	-0.02823	0.00743	0.00014	12.79214	Site_B	40.94109
5	23	771.2624	0.28464	0.09025	0.00161	1.5143	0.42217	0.00033	-0.05726	0.01571	0.00027	13.22313	Site_B	19.41989
8	23	765.9313	0.27623	0.18015	0.1252	1.81077	1.35848	0.18255	-0.06883	0.05104	0.17745	13.15346	Site_B	14.08878
11	23	756.07804	0.2785	0.10346	0.00711	2.29072	0.79251	0.00385	-0.08677	0.02945	0.00322	13.20034	Site_B	4.23552
14	23	752.45471	0.27496	0.09258	0.00298	2.51069	0.65815	0.00014	-0.09498	0.02421	0.00009	13.21643	Site_B	0.61219
17	23	752.51956	0.27403	0.09055	0.00248	2.60444	0.62835	0.00003	-0.09818	0.0229	0.00002	13.26364	Site_B	0.67704
20	23	752.2357	0.26635	0.09069	0.00331	2.81485	0.69229	0.00005	-0.10487	0.02495	0.00003	13.42093	Site_B	0.39318

23	23	753.29208	0.25875	0.0916	0.00473	2.97038	0.73377	0.00005	-0.10935	0.0262	0.00003	13.58143	Site_B	1.44957
26	23	755.98701	0.25363	0.09516	0.00769	3.04248	0.79881	0.00014	-0.1105	0.02825	0.00009	13.76709	Site_B	4.1445
29	23	760.31867	0.25231	0.10002	0.01165	3.04425	0.857	0.00038	-0.10902	0.03001	0.00028	13.96173	Site_B	8.47615
32	23	761.99131	0.25538	0.09919	0.01003	3.13477	0.83971	0.00019	-0.11117	0.02914	0.00014	14.09878	Site_B	10.14879
35	23	764.74875	0.25797	0.09812	0.00856	3.15326	0.8229	0.00013	-0.11093	0.02831	0.00009	14.21222	Site_B	12.90623
2	26	793.37631	0.31469	0.09539	0.00097	0.7181	0.19473	0.00023	-0.02809	0.00738	0.00014	12.78199	Site_B	41.53379
5	26	771.74453	0.30826	0.09935	0.00192	1.50314	0.38825	0.00011	-0.05683	0.0144	0.00008	13.22494	Site_B	19.90202
8	26	766.40058	0.29837	0.12221	0.01463	1.79784	0.68095	0.00829	-0.06836	0.02545	0.00722	13.14974	Site_B	14.55806
11	26	756.39646	0.30137	0.10465	0.00398	2.28444	0.64861	0.00043	-0.0866	0.02402	0.00031	13.18971	Site_B	4.55394
14	26	752.59519	0.29954	0.09878	0.00243	2.51336	0.58656	0.00002	-0.09516	0.02154	0.00001	13.20555	Site_B	0.75267
17	26	752.48646	0.30092	0.09783	0.0021	2.61823	0.57445	0.00001	-0.09874	0.0209	0	13.25772	Site_B	0.64394
20	26	751.84252	0.29872	0.09694	0.00206	2.85794	0.619	0	-0.10646	0.02226	0	13.42245	Site_B	0
23	26	752.86719	0.29048	0.0971	0.00278	3.03013	0.65627	0	-0.11153	0.02339	0	13.58468	Site_B	1.02467
26	26	755.8538	0.27848	0.09951	0.00514	3.09388	0.71927	0.00002	-0.11235	0.0254	0.00001	13.76907	Site_B	4.01129
29	26	760.30966	0.27302	0.1037	0.00847	3.08653	0.78039	0.00008	-0.11053	0.02729	0.00005	13.96219	Site_B	8.46715
32	26	762.04891	0.27426	0.10402	0.00837	3.17725	0.7859	0.00005	-0.11267	0.02724	0.00004	14.09969	Site_B	10.20639
35	26	764.88911	0.27569	0.10372	0.00786	3.19872	0.78045	0.00004	-0.11251	0.02682	0.00003	14.21504	Site_B	13.04659
2	29	796.60047	0.28726	0.10166	0.00472	0.70826	0.19812	0.00035	-0.02769	0.00751	0.00023	12.78753	Site_B	44.75796
5	29	775.2205	0.27949	0.16131	0.08316	1.47824	0.79293	0.06228	-0.0559	0.0296	0.05902	13.22336	Site_B	23.37798
8	29	769.50102	0.27867	0.08111	0.00059	1.77206	0.35512	0	-0.0674	0.01302	0	13.14551	Site_B	17.6585
11	29	759.30379	0.28397	0.10045	0.0047	2.25278	0.44342	0	-0.08543	0.01634	0	13.1845	Site_B	7.46127
14	29	755.20559	0.28438	0.13198	0.03119	2.48613	0.93981	0.00816	-0.09421	0.03461	0.00649	13.19395	Site_B	3.36307
17	29	755.01947	0.28554	0.11287	0.01141	2.58842	0.71502	0.00029	-0.09772	0.02605	0.00018	13.24467	Site_B	3.17695
20	29	754.22291	0.286	0.10799	0.00809	2.8343	0.71837	0.00008	-0.10564	0.02583	0.00004	13.41513	Site_B	2.38039
23	29	754.83708	0.28402	0.10659	0.00771	3.03855	0.73779	0.00004	-0.11185	0.02627	0.00002	13.58332	Site_B	2.99456
26	29	757.7127	0.27175	0.11259	0.01579	3.12163	0.87388	0.00035	-0.11337	0.03087	0.00024	13.76724	Site_B	5.87019
29	29	762.33823	0.2603	0.1365	0.05654	3.11369	1.23331	0.01158	-0.11153	0.04324	0.0099	13.95847	Site_B	10.49572
32	29	764.08342	0.25973	0.13196	0.04903	3.20435	1.17411	0.00635	-0.11366	0.04082	0.00536	14.09656	Site_B	12.2409
35	29	766.97112	0.25946	0.12587	0.03927	3.22681	1.08871	0.00304	-0.11351	0.03753	0.00249	14.21366	Site_B	15.1286
2	32	798.1306	0.27666	0.10735	0.00996	0.70985	0.20193	0.00044	-0.02776	0.00767	0.00029	12.78693	Site_B	46.28808
5	32	777.18698	0.26149	0.08727	0.00273	1.46393	0.30401	0	-0.0554	0.01117	0	13.21275	Site_B	25.34446
8	32	771.56283	0.26756	0.08821	0.00242	1.7488	0.35142	0	-0.06657	0.01291	0	13.13594	Site_B	19.72031
11	32	761.23129	0.27406	0.08922	0.00213	2.22616	0.4266	0	-0.08449	0.01557	0	13.17395	Site_B	9.38877
14	32	756.79787	0.27732	0.10887	0.01086	2.4705	0.47615	0	-0.09368	0.01748	0	13.18598	Site_B	4.95535
17	32	756.41498	0.28157	0.1405	0.04507	2.57871	0.99275	0.00939	-0.09743	0.0362	0.00712	13.2341	Site_B	4.57247
20	32	755.58367	0.28239	0.12195	0.02058	2.82273	0.84603	0.00085	-0.10528	0.03042	0.00054	13.40544	Site_B	3.74115
23	32	756.01687	0.28384	0.11707	0.01533	3.04217	0.82171	0.00021	-0.11202	0.02925	0.00013	13.57871	Site_B	4.17435
26	32	758.58101	0.27646	0.12614	0.0284	3.14987	1.02023	0.00202	-0.11442	0.03603	0.00149	13.76417	Site_B	6.7385

29	32	763.2111	0.26268	0.27862	0.34579	3.14582	3.01223	0.29632	-0.11273	0.10571	0.28626	13.95328	Site_B	11.36858
32	32	765.07214	0.25813	0.26271	0.32582	3.23383	2.84373	0.25546	-0.11475	0.09905	0.24668	14.0913	Site_B	13.22962
35	32	767.9503	0.25701	0.17557	0.14324	3.25654	1.75587	0.06364	-0.11459	0.06066	0.05888	14.21011	Site_B	16.10778
2	35	799.41947	0.26341	0.11089	0.01753	0.71518	0.20709	0.00055	-0.02794	0.00786	0.00038	12.79774	Site_B	47.57695
5	35	778.42096	0.25321	0.09224	0.00605	1.46518	0.30357	0	-0.05548	0.01117	0	13.20395	Site_B	26.57844
8	35	772.96678	0.25748	0.09375	0.00603	1.74171	0.34897	0	-0.06635	0.01284	0	13.12474	Site_B	21.12426
11	35	762.76583	0.26356	0.09558	0.00583	2.21078	0.42248	0	-0.08398	0.01545	0	13.16258	Site_B	10.92331
14	35	758.2436	0.2681	0.09606	0.00526	2.45412	0.45609	0	-0.09313	0.01659	0	13.17559	Site_B	6.40108
17	35	757.71566	0.27202	0.1159	0.01893	2.5643	0.49491	0	-0.09695	0.01805	0	13.22483	Site_B	5.87314
20	35	756.80521	0.27477	0.16673	0.09935	2.80904	1.41174	0.04662	-0.10486	0.05079	0.03896	13.39468	Site_B	4.96269
23	35	757.24057	0.27587	0.13726	0.04445	3.02765	1.09436	0.00566	-0.11156	0.03896	0.00419	13.5694	Site_B	5.39805
26	35	759.63684	0.27157	0.19342	0.16032	3.15111	1.99595	0.11439	-0.11452	0.0705	0.1043	13.75809	Site_B	7.79432
29	35	764.05935	0.26057	0.11299	0.0211	3.16278	0.59974	0	-0.11339	0.02092	0	13.94599	Site_B	12.21683
32	35	765.92986	0.25439	0.11255	0.02381	3.25339	0.61268	0	-0.11552	0.02119	0	14.08204	Site_B	14.08735
35	35	768.90084	0.24993	0.112	0.02565	3.27467	0.61979	0	-0.11529	0.02126	0	14.20148	Site_B	17.05832